Supplementary information: EASI estimation alongside clinical targets and serum trajectories in a Chinese atopic dermatitis cohort

Supplementary guide

The Supplementary Methods define the analysis populations, endpoints and validation. Tables S1–S16 provide exact target rates, complete model metrics, safety-documentation coverage, reference-exclusion sensitivity, input/reference specifications, matched input comparisons, IgE missing-data sensitivity, laboratory coverage, IGA-definition sensitivity classification intervals, baseline serum concentrations, framework comparisons, molecular model extensions, transport benchmarks and single-visit low-activity states. Figures are grouped by their clinical or measurement question below. Complete aggregate results, including negative and non-estimable findings, remain in the linked CSV collection. Missing estimates are not zero effects. **Cohort structure and treatment comparability**. Clinical and model eligibility, baseline balance and common treatment support.

Clinical outcomes and target comparisons. Observed and adjusted clinical scores, laboratory follow-up and all specified treatment contrasts.

Serum changes and regional disease. All available-pair molecular windows, complete adjusted treatment contrasts, clinical associations and regional burden. Fixed three-visit courses are in main Figure 4; baseline concentrations are in Table S12.

Individual EASI estimation and monitoring. Observed-versus-estimated scores, paired changes and response classification. Severity and holdout estimates remain in numerical source tables.

dupi denotes dupilumab; abro denotes abrocitinib, with initial dose stated in mg. Treatment colours are fixed throughout. Molecular changes use $\log_2$ concentration units; a change of -1 means a halving. Model errors use EASI points. Correlation describes ordering, whereas absolute errors and concordance address numerical agreement.

1 Supplementary Methods

1.1 Study design and clinical records

This analysis used longitudinal records from a Chinese observational treat-to-target project. The supplied protocol specified clinician-directed systemic treatment rather than random allocation. We retained the initial treatment group for longitudinal comparisons and separately extracted subsequent dose, interval, stopping, switching and restart information. The analysed records came from CAMS and Huashan. The protocol's planned centre list and planned sample size were not substituted for the observed cohort.

The clinical dose-comparison population included patients aged at least 12 years with a known initial regimen. Six of 170 source patients were below age 12 and one had an unresolved initial dose, leaving 163 patients (65/56/42 by initial treatment). Records with unresolved simultaneous initial-dose assignments were excluded from dose-specific comparisons but retained in the source audit. Adult DLQI analyses used age at least 18. Baseline Table 1 uses one observed W0 record per eligible patient, with no weighting or imputation. Continuous summaries use median and quartiles with linear interpolation, rank $1+(n-1)p$ (type 7; Excel PERCENTILE.INC); all available denominators are reported. Age and BMI are also presented as medians, rather than means and standard deviations. Repeated records were linked by the study patient identifier and nominal visit; a patient was never treated as a new independent person at each visit. The requested reporting visits were W0, W2, W4, W12 and W24 for PP-NRS; W0, W4, W12 and W24 for EASI, IGA, DLQI, BSA, SCORAD and eosinophils; and W0, W12 and W24 for IgE. Low-coverage intermediate visits were retained in the underlying data but were not inserted into these trajectories.

EASI was the recorded clinical reference outcome. Source discrepancies between regional sums and total EASI were audited rather than silently repaired. IGA grades were retained as recorded, including grade 5. Confirmation of the exact IGA version and PP-NRS recall window remains necessary. Nominal clinical/serum visit linkage does not by itself verify that the blood draw and examination occurred on the same calendar day; exact dates and a permissible matching window require source confirmation. Among the 152/89/75 model visits at W0/W12/W24, 8/2/0 had complete regional scores whose sum differed from total EASI by more than 0.15; the largest absolute gaps were 18.7/8.5/0.0 points, and 1/1/4 visits lacked a complete regional reference. Restricting evaluation to all four regions observed with total–sum difference at most 0.01 retained 140/86/71 visits and gave MAEs 3.17/1.32/1.11. This evaluation-only sensitivity uses unchanged saved predictions. A separate training-exclusion sensitivity removed the eight W0 and two W12 discrepant visits from fitting, retained missing-region visits, and repeated preprocessing and inner structure/penalty selection within the original patient folds. Evaluation patients and recorded outcomes were unchanged. It gave W0/W12/W24 MAEs of 3.19/1.15/1.06; matched BSA controls were refitted identically. Neither analysis adjudicates which clinical reference is correct (Supplementary Table S4). Details are in `model_reference_QC_audit.csv`.

The supplied protocol is internally inconsistent on IGA: its eligibility text states 0–4, whereas Appendix 5 defines 0–5, with grade 5 labelled very severe. All 22 recorded grade-5 values occurred at baseline (9/2/11 by initial treatment) and were retained. A post hoc definition sensitivity removed the IGA alternative from the physician domain, leaving EASI50 at W4, EASI75 at W12 and EASI75 or EASI$\leq$ 5 at W24; patient-domain rules were unchanged. Both definitions were compared on the same jointly evaluable patients, and their own available-case denominators were also reported. Three-valued AND/OR logic preserved unknown components. An additional sensitivity excluded the 22 patients with baseline IGA5; this was a different population, not a correction of known errors. Complete paired transitions and patient-deduplicated changes are supplied in the IGA-definition source tables and Supplementary Table S10. The protocol also specified 24-hour peak itch recall, without confirming its consistent implementation in the registry.

1.2 Local framework and comparison with published targets

The project presentation describes evidence review, stakeholder consultation, a questionnaire on assessment instruments, feedback and discussion, a second questionnaire on targets and timing, and steering-group review. The planned stakeholder sample comprised 30 physicians, 10 nurses and 10 patients; 49 returned questionnaires were reported. The planned agreement threshold was at least 75%. These source descriptions do not provide individual ballots, round-specific denominators or item-level agreement. The flow diagram therefore represents the documented process, without assigning unverified voting results. Later month-9/12 targets in the source framework were not evaluated as observed outcomes in this 24-week cohort.

Supplementary Table S13 compares the local framework with international 2021, Chinese 2022, Canadian 2023, AHEAD 2024, STRIVE AD and IEC 2026 recommendations. Original English publications were checked; the Chinese guidance thresholds were cross-checked against the guidance authors' 2026 clinical-trial analysis.[1–7] The comparison is descriptive and does not retrospectively redefine the study endpoints. AHEAD minimal disease activity depends on agreed optimal targets. Using its EASI branch, IEC low disease activity requires EASI $\leq$ 7 and PP-NRS $\leq$ 4, and very low disease activity requires EASI $\leq$ 3 and PP-NRS $\leq$ 1. IEC complete control

and off-drug remission additionally require persistence for at least six and twelve months, respectively. These duration-defined states were not inferred from isolated visits. Recorded local IGA was not silently recoded as vIGA-AD.

1.3 Clinical targets and treatment comparisons

The week-2 target was PP-NRS ≤ 4. At week 4, the physician domain was EASI50 or an IGA decrease of at least two points, and the patient domain was PP-NRS ≤ 4 or adult DLQI ≤ 10. At week 12, the physician domain was EASI75 or IGA 0/1, and the patient domain was PP-NRS 0/1 or adult DLQI ≤ 4. At week 24, EASI ≤ 5 was an additional physician-domain criterion. The combined target required both domains. Because instrument identity and recall windows remain unresolved, these are exploratory targets based on the recorded scores. These are the investigators' operational definitions; they were not retrospectively relabelled as the newer IEC low-disease-activity or remission definitions.[7]

For an OR criterion, one observed positive component established success; a negative result required all components to be observed and negative. For an AND criterion, one observed negative component established failure; success required both positive components. Other cases remained unknown. Adult DLQI was an unknown component for ages 12–17, not a negative component and not removed from the OR rule. Thus adolescent itch success establishes the patient target; itch failure alone does not establish patient-target failure. The audit exhaustively verified the nine two-component truth states and reconciled every saved composite. Among 15 adolescents, the patient domain had 6/0/9 successes/failures/unknowns at W4, 4/0/11 at W12 and 4/0/11 at W24; the combined domain had 5/1/9, 4/5/6 and 4/3/8. An explicitly different PP-NRS-only adolescent rule would make five W12 and four W24 patient-domain unknowns determinate, including one W24 combined outcome. This is a sensitivity count, not a redefinition of the primary composite. Age-stratified counts are in `composite_age_evaluability_audit.csv`. EASI response required an observed positive baseline. Individual thresholds were also analysed separately. Proportions used endpoint-specific observed denominators and Wilson 95% confidence intervals. Overall three-group tests used conditional Monte Carlo Pearson tests with 49,999 draws, and pairwise comparisons used Fisher's exact test and risk-difference intervals. The three pairwise comparisons within an endpoint and visit formed a Holm family; exploratory FDR results were also retained. Overall tests were not included as an extra member of that pairwise family.

Baseline-adjusted binary target rates and risk differences used generalized overlap weights from a multinomial initial-treatment model with baseline EASI, age, sex, BMI and centre. We used 1,000 patient bootstrap resamples, stratified by initial treatment and refitting the propensity model, for percentile intervals. Two-sided P values used normal Wald tests based on bootstrap standard errors; three-group tests used a two-degree-of-freedom Wald statistic. The three individual pairwise comparisons formed a Holm family. Weighted results supplement the observed numerator and denominator. The primary target analysis used bootstrap seed 20261017. A revision W2 sensitivity added baseline PP-NRS to the propensity model. One patient lacked baseline PP-NRS; both the original-covariate model and the added-itch model were fitted on the remaining 162 patients, with the original 163-patient analysis retained. The same standardized logistic model (ridge parameter $C = 1$), 1,000 stratified patient resamples and seed 20261018 were used for this sensitivity. The revision files also contain a separate full-cohort run with seed 20261018. Its point estimates use the same analysis, but its Monte Carlo intervals and Wald P values are not the frozen primary results; borderline changes were not selected for the main claim. The added-itch 100 mg–dupilumab contrast was 41.4 percentage points (95% CI, 13.0–66.8; three-contrast Holm $P = 0.0104$); the 100-minus-200 mg contrast was 6.7 points (-11.3 to 26.2). The largest endpoint-specific absolute standardized difference was 0.53 even after adding itch, so residual imbalance remained. Baseline weights were retained when calculating available-case balance and ESS; no claim of balanced follow-up selection is made. The table below reports both unweighted and matching baseline-overlap-weighted worst/best bounds, assigning every unknown outcome failure/success while retaining baseline weights. These bound the initial cohort under extreme missing-outcome assignments; the reported weighted available-case rate represents patients with that endpoint observed. Endpoint-specific overlap-weight ESS is reported separately. Bounds are sensitivity limits, not confidence intervals.

Unweighted observed outcomes and initial-cohort bounds

Target	Initial group	Success/evaluable	Initial N	Observed rate, %	Unweighted missing-outcome bounds, %
W2 itch	dupi	15/27	65	55.6	23.1–81.5
W2 itch	abro 100 mg	34/37	56	91.9	60.7–94.6
W2 itch	abro 200 mg	33/36	42	91.7	78.6–92.9
W24 combined	dupi	26/36	65	72.2	40.0–84.6
W24 combined	abro 100 mg	22/24	56	91.7	39.3–96.4
W24 combined	abro 200 mg	20/23	42	87.0	47.6–92.9

Matching baseline-overlap-weighted analysis

Target	Initial group	Weighted available-case rate, %	Available-case weight ESS	Weighted missing-outcome bounds, %
W2 itch	dupi	51.9	17.6	19.4–82.0
W2 itch	abro 100 mg	93.6	25.1	76.0–94.8
W2 itch	abro 200 mg	85.8	21.1	63.3–89.5
W24 combined	dupi	46.2	20.8	22.0–74.3
W24 combined	abro 100 mg	85.4	14.7	26.2–95.5
W24 combined	abro 200 mg	87.0	12.5	38.3–94.3

Bounds assign all unknown outcomes failure or success in the initial cohort. Weighted bounds retain the same baseline overlap weights in observed and unknown records. Weighted available-case rates use only endpoint-observed patients. ESS is the effective sample size of those weighted available cases, not a patient count or an incidence denominator. Bounds are sensitivity limits, not confidence intervals.

For generalized propensity scores $e_t(x)$, the weight is $h(x)/e_t(x)$, with $h(x) = 1/\sum_t[1/e_t(x)]$. The baseline weighting function defines a common treatment-overlap population; an available-outcome analysis can depart from that population through selective observation. Covariate balance, propensity distributions and analysis-specific effective sample sizes are reported; weighting does not establish elimination of confounding.[8]

Continuous treatment comparisons used marginal regression with baseline outcome, treatment-by-visit terms, age, sex, BMI and centre. Generalized overlap weights addressed measured baseline differences in initial treatment selection. Patient-cluster robust covariance accounted for repeated outcomes conditional on the estimated baseline weights, without propagating their estimation uncertainty. A singular-value decomposition check prevented reporting non-estimable treatment-by-visit cells. Exploratory contrasts of treatment differences over time used the same fitted coefficients and full patient-cluster covariance, comparing W2 to W12/W24 for itch and W4 to W12/W24 for EASI. Holm correction covered the three treatment contrasts within each endpoint and interval; these tests did not infer a time interaction from different visit-specific significance levels. IgE and eosinophils used $\log(1+x)$; back-transformed estimates are not arithmetic means. No dupilumab W4 eosinophil measurements were available. That treatment-by-visit cell was not estimable without additional cross-treatment or temporal restrictions, so it remained blank. Multiple imputation would require unsupported assumptions about that unobserved cell and was not used to complete the observed trajectory. The fixed-patient IgE display uses the same patients at W0/W12/W24; adjusted comparisons are shown separately from raw medians. Eosinophil paired changes use all observed pairs for each interval, with descriptive t-based 95% intervals on the recorded scale; these intervals are not multiplicity-adjusted treatment-comparison tests. Supplementary Table S9 reports laboratory coverage and these paired changes. Unweighted adjustment, group balance, observation patterns and supplementary dupilumab-versus-pooled-abrocitinib contrasts were retained. Pooling used the observed dose mixture and did not erase the two initial abrocitinib groups. The comparisons estimate adjusted observational associations, not randomized fixed-dose effects. Nominal eligibility based on the registry filename date placed all initial patients beyond W24, but this administrative sensitivity is not an approved database lock or verified visit window. Reasons for absent assessments were unavailable.

Treatment reductions were assigned from recorded regimen intervals, with assessment/decision ordering marked uncertain at shared visit boundaries. A later reduction was not moved to an earlier visit. W12 response and subsequent regimen changes defined descriptive maintenance pathways. Analyses used observed W24 EASI75, explicit missing-outcome bounds and fixed-patient trajectories where all three visits were observed. This approach distinguishes treatment initiation from the selected maintenance pathways, consistent with the need to specify how intercurrent treatment changes enter the question being estimated.[9] It does not identify a counterfactual outcome under a different regimen. At W0, two source records labelled initial 100 mg explicitly listed a 200 mg starting regimen, and one labelled initial 200 mg listed 100 mg. These are conflicts between the source group and regimen fields, not demonstrated post-baseline switches. The group labels were retained, and `exposure_W0_source_adjudication.csv` preserves their counts separately from unknown exposure. Clinical-record adjudication is still required.

1.4 Serum measurements and molecular analyses

The laboratory source reports identify serum samples analysed with Luminex and MSD platforms. Luminex used an R&D Systems LXSAHM-13 panel and Luminex-200; the MSD reports identify K15067M-1/K15067M-2 kits and an SQ120 instrument. Reports describe 50-μL serum and twofold dilution. The service-batch sample count was not used as the study sample size. Assay lots, run identifiers and source links remain in the restricted provenance records.

The analysis used the numeric integration export rather than the vendor's filtered display matrix. Positive export numerics recorded in pg/mL were $\log_2$ transformed once, unchanged; the analysis applied no additional twofold dilution correction. This reproduces the fitted scale rather than verifying how the laboratory corrected its export for dilution. Explicit censoring strings and nonpositive values were not converted to measured positive concentrations.

The vendor's filtered output can replace values below range with zero and values above range with an upper bound; those conventions must not be interpreted as exact measured concentrations. A sensitivity analysis represented IL-5 values below twice the software calibration lower bound at that boundary and repeated the model workflow. The software boundary is not an independently validated clinical quantification limit. Cross-platform bridging, technical-repeat precision and lot effects require additional assay documentation.

All 23 measured markers were retained. Molecular changes were formed within each patient and visit pair before group summaries, avoiding subtraction of averages from different patient sets. We analysed W0–W12, W12–W24 and W0–W24 separately. Mean $\log_2$ changes were accompanied by t-based confidence intervals when at least five pairs were available. Smaller sets were descriptive. Two-sided paired mean-change sign-flip tests retained zero changes and ties. They enumerated all 2^n signs when $n \leq 15$; otherwise they used 19,999 random signs (seed 20261002), counting exceedances at the observed absolute mean with a plus-one numerator and denominator. All-zero changes gave $P = 1$. The sign-flip null assumes independent patients and sign-exchangeable paired changes (symmetry about zero); it is not a randomized-treatment test. Benjamini–Hochberg adjustment covered the full 23-marker, three-treatment, three-window family: 207 planned cells, of which 193 had estimable P values. Non-estimable cells remained missing and were not counted as tests. Between-treatment change models adjusted for starting concentration, baseline EASI, age, sex, BMI and centre, with HC3 covariance. Centre-specific and assay-batch sensitivity analyses were retained. Each between-treatment molecular correction family included all marker, window and pairwise contrasts within one centre scope and analysis variant; scopes and variants were not pooled. Molecular–clinical change correlations formed their own panel-wide family across markers, clinical outcomes, treatments and windows, and response-depth contrasts formed a separate family. The global molecular treatment-by-visit family had 23 planned marker tests, with 21 estimable. A source-table inventory records the defined families and finite-test counts. Spearman correlations related molecular changes to clinical changes within treatment groups; baseline regional correlations were assessed separately. Sample sizes, intervals and multiplicity families accompany each source table.

Regional analyses used source-consistent regional totals. Regional EASI contributions were divided by their theoretical maxima (7.2, 14.4, 21.6 and 28.8 for head/neck, upper limbs, trunk and lower limbs). Percentage improvement required a positive regional starting value; zero-start exclusions were counted. Because very low W12 values can produce extreme W12–W24 percentage changes, the corresponding figure instead shows change as a percentage of the theoretical regional maximum. Both definitions are retained in the tables. Regional treatment contrasts used unadjusted Welch comparisons and HC3 models adjusted for starting regional severity, baseline total EASI, age, sex, BMI and centre, with FDR over the 48 region-by-window-by-contrast tests in each analysis/outcome family. Whole-body PP-NRS change was not labelled as regional itch.

1.5 Fixed-patient three-visit molecular display

For each marker and initial treatment, the main molecular figure retained patients with positive recorded concentrations at all three visits. Each patient's baseline $\log_2$ concentration was subtracted from their follow-up value before averaging; baseline was zero by construction. Pointwise Student-t 95% intervals and two-sided sign-flip tests required at least five patients. All signs were enumerated for $n \leq 15$; larger sets used 49,999 random sign assignments with a plus-one correction. Tests with provisional q values between 0.04 and 0.06 were refined by exact enumeration for $n \leq 24$, or 1,999,999 random assignments otherwise, before the same family was recomputed. This addressed simulation precision without changing the hypotheses or populations.

The Benjamini–Hochberg family comprised 138 planned comparisons (23 markers, three initial treatments, two follow-ups). The 14 non-estimable hypotheses contributed $P = 1$ to the correction but had no displayed q value. Stars identify within-group changes with q$<$ 0.05; they do not test treatment differences. The previous all-available-pair family remained separate because its populations differ. Baseline summaries in Supplementary Table S12 were calculated from positive recorded concentrations on the original pg/mL numeric scale, using type-7 quartiles and marker-specific denominators. Censored and other unavailable assay states were counted separately rather than substituted into these distributions.

1.6 Common EASI estimator and validation

Eligibility required age at least 12, observed EASI and BSA, and observed positive CCL17, CCL26 and IL-5 concentrations. Ambiguous initial-dose labels did not exclude otherwise eligible model records, because treatment was not a required input. The three-marker panel arose from earlier exploratory work and was fixed for this revision. No clinical outcome was imputed. Model development used W0 and W12; W24 did not enter fitting or selection. Because earlier project analyses had already examined W24, its results are described as within-cohort temporal application rather than untouched external validation.

All observations from one patient remained in the same outer fold. Five-fold validation was repeated twice using the established patient partitions. Each training fold contained development visits only. Three-fold patient-level inner validation selected the ridge penalty from 0.1, 1, 10 and 100, minimizing mean per-patient absolute error. Each patient received equal total fitting weight. Feature centring and scaling used training data only. Three common structures were considered: an additive BSA-plus-three-marker model; the same model with quadratic BSA; and BSA multiplied by a linear molecular component without a free intercept. Structure and penalty selection were nested together for the primary performance estimate. No treatment, phase, time, previous score, age or BMI was required at deployment.

After full-development inner selection, the fitted common equation was

$$\widehat{\mathrm{EASI}} = \mathrm{clip}_{[0,72]}\{\mathrm{BSA}\,[0.512721579 + 0.009847080\log_2(\mathrm{CCL17}) + 0.009646929\log_2(\mathrm{CCL26}) - 0.008270558\log_2(\text{IL-5})]\}.$$

BSA is entered as a percentage from 0 to 100, and concentrations must use the development export's recorded pg/mL numeric scale, with $\log_2$ applied once. The implementation rejects missing, nonfinite or nonpositive molecular inputs and invalid BSA values. The area constraint returns zero when BSA is zero. It is a statistical constraint reflecting absent observed lesion extent, not proof of absent molecular inflammation. The released parameter file retains full precision. The main-text display multiplies the molecular coefficients by 100 and uses BSA/100, allowing two decimal places without dropping the smaller molecular terms. An explicit rounding audit gave nested-selection W0/W12/W24 MAEs of 3.194/1.298/1.193, compared with 3.194/1.297/1.192 at full precision. All primary results use full precision; the rounded equation is a reading aid. For the final fitted equation, maximum absolute arithmetic differences were 0.075550, 0.024807 and 0.018474 EASI points at W0/W12/W24; these are rounding differences, not clinical prediction errors. Response classifications at EASI50/75/90 were unchanged in the audit, but one of 150 week-24 fold-level evaluations crossed the absolute EASI 7 threshold, so threshold-based use should retain full precision. This single final fit has not received independent validation; primary metrics evaluate the nested development procedure that selected it.

Comparators used the same patients and folds: BSA alone, the three markers alone, the additive common model, and the earlier stage-interaction model. We evaluated MAE, RMSE, signed error, Spearman correlation, Lin's concordance correlation, R^2, calibration intercept and slope, severity-band agreement and proportions within one and two EASI points. Metrics were calculated within each repetition before averaging, not from averaged predictions. Patient bootstrap intervals used 2,000 resamples of held-out errors. The same sampled patient indices were used in every repetition and comparator; metrics were scored within each repetition before averaging. These intervals describe uncertainty conditional on the saved cross-validation predictions: neither nested selection nor fitting was repeated, and earlier adaptive marker exploration is not covered. Calibration intervals resampled patients in the regression of observed on predicted EASI. The primary nested-procedure minus area-only MAE differences were 0.12 (95% CI, 0.04–0.20) at W12 and 0.17 (0.11–0.24) at W24. These paired estimates differ from the earlier fixed-structure comparison and do not establish molecular added accuracy. All comparator and fixed-structure results remain in the source tables. Descriptive agreement limits were mean error plus or minus 1.96 error standard deviations.[10]

Severity-band agreement used four categories: EASI ≤ 7, > 7 to ≤ 21, > 21 to ≤ 50 and > 50. Error subgroup analyses instead combined the two highest categories, using three strata: EASI ≤ 7, > 7 to ≤ 21 and > 21. Binary sensitivity and specificity used EASI > 7 and are not interchangeable with every original p-EASI classification definition. Leave-centre-out and leave-initial-treatment-group-out analyses removed all records from held-out patients. Changes were paired within patient, repetition and fold before averaging. Each of W0–W12, W0–W24 and W12–W24 was evaluated separately; the windows share patients. EASI50/75/90 classification required positive observed and estimated starting scores, with exclusions counted. Observed response used observed scores at both visits, and estimated response used estimated scores at both visits. Scatter plots and any integer confusion matrices use the first outer repetition; the primary response forest and principal metrics average repetition-specific values. They must not be interpreted as one common confusion matrix. EASI75 sensitivity used 49 and 54 observed responders at W12 and W24; specificity used 36 and 13 observed nonresponders, respectively, in both repetitions. For sensitivity, specificity and overall agreement, each repetition is reported with a Wilson 95% binomial interval. For the repetition-average estimate, Clopper–Pearson intervals use a Bonferroni level of $1 - 0.05/R$ for each of R repetitions, and their lower and upper limits are averaged separately. Within each endpoint, window and metric, these give conservative simultaneous bounds across repetitions under a conditional binomial working model without assuming independence between repetitions. They do not establish exact coverage for the full cross-validation procedure, whose shared fitted models and development uncertainty are not resampled. Patients are not multiplied by the number of repetitions. Repetition-level counts and intervals are in Supplementary Table S11; average bounds are in `primary_EASI75_uncertainty.csv`. Earlier percentile-bootstrap limits are retained as explicitly labelled diagnostic columns, not the primary classification bounds. The empirical cumulative absolute-error curve is the average of the two repetition-specific proportions. Pointwise bootstrap intervals carry all saved predictions from each sampled patient together and do not refit the development procedure. Applying the published formulas

used the published rounded coefficients without local refitting, with local IL2RA as a nominal sIL-2R proxy and identical eligible patients for both methods. Original p-EASI outputs were initially left unbounded to preserve the published arithmetic, whereas local estimates were clipped to 0–72; matched clipping of the original outputs was examined separately. Patient-paired bootstrap intervals were calculated for differences between the current and published-formula metrics, retaining both repetitions for each sampled patient. These intervals characterize the local comparison; they do not remove the difference in inputs or establish assay equivalence.[11–13]

1.7 Independent auxiliary target models

PP-NRS and adult DLQI were fitted separately with original-scale or $\log(1+x)$ ridge regression, using patient-level inner validation to select the link and penalty. Predictions were bounded to 0–10 and 0–30, respectively. Recorded IGA used a penalized proportional-odds model over the six observed grades, with ordered thresholds and six category probabilities. Normalized ranked probability score was the primary ordinal loss. All three targets used the same four-input panel for a practical shared measurement set, with BSA-only, molecular-only and training-only constant comparators. Neither EASI nor the target's current observed score was an input. Development and temporal application followed the patient-level W0/W12 and W24 separation used above. This limited shared-panel study does not claim to select the optimal molecular set independently for each target. Details and full estimates are in the supplement.

1.8 Reproducibility and reporting

The model analysis used Python 3.9.6, NumPy 2.0.2, pandas 2.3.3, SciPy 1.13.1 and scikit-learn 1.6.1. Exact dependency versions, analysis specifications, input/output hashes and execution logs accompany the code. Clinical and prediction reporting follow the relevant STROBE and TRIPOD+AI domains; applicability and risk-of-bias issues are recorded explicitly.[14–16] No non-significant difference, high correlation or overlapping mean trajectory was taken as proof of individual interchangeability.

Matched input ablation

The original core population contained 316 visits from 160 patients; current PP-NRS was available for 315 visits from the same 160 patients. Adult current-itch and DLQI comparisons contained 207 visits from 111 patients. Models within each population used identical observations and the original patient outer folds. Development comprised W0/W12 only: 241 visits from 156 patients in the core, 240 visits from 156 in the itch comparison, and 163 visits from 107 in the adult comparison. Five outer folds with two repetitions and three inner patient folds selected ridge penalty and, for BSA-containing candidates, linear, BSA-quadratic or BSA-times-other-input structures. Serum-only and itch-plus-serum models used linear ridge. Standardization was fitted inside each training partition and each patient's total fitting weight was equal. Predictions were bounded to 0–72. No clinical outcome or deployment input was imputed.

CCL17-only, BSA-plus-CCL17 and fixed-area leave-one-marker-out comparisons tested panel dependence. Coefficients were summarized only within the same structure; different parameterizations were not averaged. All metrics were calculated within each repetition before averaging. Paired differences resampled the same patients across models and repetitions in 2,000 draws, conditional on saved predictions. These post-result exploratory comparisons did not replace the frozen formula and did not correct for the earlier panel selection. Complete metrics, paired changes, coefficients and selection frequencies accompany Tables S6–S7.

IgE observation patterns and multiple imputation

IgE was observed at W0, W12 and W24, with no W4 measurement. All available two-visit pairs were analysed without requiring the third visit. Paired t intervals used log(1+IgE), so exponentiated mean differences are geometric mean ratios of IgE+1, not arithmetic mean ratios. An available-data maximum-likelihood random-intercept model included treatment-by-visit, baseline EASI, age, sex, BMI and centre where variable, with unconstrained treatment-specific baseline means. CAMS (130 patients) supplied observed follow-up in every initial treatment; Huashan supplied no dupi IgE at W12 or W24. Full-cohort imputation of 163 patients therefore involved unsupported treatment–centre follow-up cells and was retained only as an additional sensitivity. Five CAMS patients, and 16 across both centres, had no IgE at any of the three visits; their imputations relied entirely on clinical covariates and relationships learned from other patients.

Fully conditional imputation used one wide record per patient. Four chains each underwent 200 burn-in cycles and retained 100 draws separated by ten cycles. Incomplete variables were the three IgE values and W12/W24 EASI; complete predictors were initial treatment, baseline EASI, age, sex, BMI and centre where variable. Future

observed measurements served as auxiliaries for retrospective reconstruction, not prospective prediction. EASI auxiliary imputations were used only inside the imputation model, never as validation outcomes. Type-1 predictive mean matching used five closest observed same-treatment/same-visit donors on log(1+IgE). The alternative drew coefficients, residual variance and positive IgE values from Bayesian Gaussian regression on log(IgE), without an invented upper ceiling. Both assumed common conditional slopes across treatment groups with separate intercepts. Observed values were unchanged.

Rubin variance and Barnard–Rubin small-sample degrees of freedom pooled group means, changes and baseline-adjusted contrasts. The latter used baseline log(1+IgE), treatment-by-visit, baseline EASI, age, sex, BMI and centre, with fixed previously estimated baseline overlap weights and patient-cluster covariance. They condition on those weights and do not include propensity-estimation uncertainty. Initial groups were not reassigned after treatment changes. Illustrative MNAR scenarios multiplied only missing follow-up IgE+1 by one-half or two, retaining a zero lower bound for IgE. The repeated value 5000 was not assumed to be a verified censoring limit. Mixing, effective draws, missing-information fractions, Monte Carlo errors, donor support and predictive tails were checked. The maximum MAR change diagnostic was $\widehat{R} = 1.013$, with Monte Carlo error at most 5.6% of the inferential standard error. In CAMS abro 200 mg at W24, 13 donors supported 28 missing values; 25.8% of PMM recipient fitted means lay outside the donors' fitted-mean range, averaged across imputations. Gaussian draws could extrapolate, but 7.6% of imputed W24 values in that group exceeded the recorded maximum of 5000. Both methods were retained rather than selected by the direction of change; computational convergence did not remove this sensitivity.[17, 18]

Eosinophil interpolation sensitivity

No dupi patient had an observed W4 eosinophil value. Within the 15 patients with W0 and W12 measurements, the estimated value was $\exp\{(2/3)\log(1+\mathrm{EOS}_0)+(1/3)\log(1+\mathrm{EOS}_{12})\}-1$. The group median and a conditional 95% percentile interval used 2,000 patient bootstrap samples. A seven-patient subset observed at W0/W12/W24 was analysed separately; its estimated median was 0.396 (0.183–0.455), illustrating the effect of changing the population. Neither subset was spliced into an all-patient observed trajectory without identifying the denominator.

Sensitivity scenarios used raw-scale linear, early logarithmic-time and delayed quadratic-time interpolation. All assumed a smooth path and therefore did not encompass a transient rise. Masking observed W4 values in abro 100/200 mg supplied limited back-checks: log-scale interpolation MAEs were 0.162/0.222 recorded units in 22/11 patients. These checks were not validation of the missing dupi W4 measurements. No estimated EOS value entered target-rate calculations, treatment-effect inference or clinical prediction validation (Supplementary Fig. S6).

Exploratory molecular-only model extension

This extension used the original patient partitions and the core 316 visits from 160 patients (W0/W12/W24: 152/89/75). Development comprised 241 W0/W12 visits from 156 patients. Two repeats of five outer patient folds and three inner patient folds separated fitting, selection and evaluation. W24 did not enter fitting or tuning. Because this cohort had already been used extensively for model exploration, the comparison was exploratory and did not reset the validation history.

The compact panel contained CCL17, CCL26 and IL-5. The expanded panel contained the 19 markers already designated eligible for model discovery; CCL18, IL-13, IL-22 and IL-31 remained in descriptive analyses but were not promoted into the deployment panel because their earlier measurement limitations remained. Missing molecular values were replaced by the training-fold median and standardized using training-fold parameters. No clinical outcome was imputed and no missingness indicator was used. The models took only current molecular measurements at prediction, without BSA, clinical history, symptoms, treatment, time, centre or demographic inputs. Patient identity aligned folds and training pairs; recorded IGA served only as an auxiliary training outcome.

Thirty-four fixed candidates per panel comprised four weighted ridge models, 24 nonlinear radial-basis candidates and six auxiliary-target reduced-rank candidates. Ridge penalties were 0.1, 1, 10 and 100, with an unpenalized intercept and equal total fitting weight per patient. The nonlinear family used a 32-component Nyström approximation, two kernel widths, three ridge penalties and paired-change weights of 0, 0.3, 1 and 3. Its loss added a penalty on disagreement between observed W0–W12 EASI change and the difference between the corresponding model estimates. A zero weight allowed nonlinear learning without the paired constraint. This training loss did not require a baseline sample at deployment.

The auxiliary-target family jointly fitted standardized EASI and five ordered indicators, IGA$\geq$ 1 through IGA$\geq$ 5, with the IGA block scaled to comparable total weight. A rank-one or rank-two projection of training fitted responses was then used; only the EASI output was retained for prediction. The procedure did not assume that IGA was continuous or validate its recorded scale. Inner-fold mean per-patient absolute EASI error selected the

candidate, and predictions were bounded to 0–72. Family-specific and overall-selection results were both preserved (Supplementary Table S14).

A separate expanded-data sensitivity removed the BSA and three-marker-completeness restrictions, requiring observed EASI, age at least 12, a linked blood sample and at least eight observed markers from the 19-marker panel. It retained 327 visits from 161 patients (155/94/78), with existing patients retaining their original folds. Development used 249 W0/W12 visits from 157 patients. Its 19-marker selection MAEs were 7.51/7.44/8.23; these were not directly substituted for the matched-population comparison. Intervals used 2,000 patient resamples of saved held-out predictions, with identical patients resampled across comparators and repetitions. These intervals condition on fitted predictions and do not include the full history of model adaptation. The primary E-EASI formula and parameters were unchanged.

Retrospective molecular formula-transfer benchmark

Existing molecular-only equations fitted to local W0/W12 data were applied unchanged to OMIX002850; these were separate serum-only ablation models, not E-EASI with BSA deleted. Eligible observations required measured EASI between 0 and 72 and positive finite CCL17, CCL26, IL-5, IL-22 and the recorded soluble IL-2 receptor proxy. Healthy controls were excluded, and no missing input or outcome was imputed. The matched public-data analysis included 125 baseline and 60 W16 observations.[19]

The primary p-EASI comparator used the 2017 full-precision pretreatment/during-treatment coefficients; published rounded coefficients were a sensitivity. All methods were bounded to 0–72 identically for the primary comparison, with unbounded outputs retained separately. MAE, RMSE, signed error, Spearman correlation, concordance, calibration and error-tolerance proportions were calculated within visit. Changes were formed only after pairing the same 60 patients. Patient bootstrap intervals and paired MAE differences used 2,000 identical resamples across methods (seed 20261101). No model was fitted, tuned or recalibrated against the public-data EASI values in this benchmark.

A separate local comparison required all inputs for the molecular formulas and BSA, retaining 60/12/13 visits at W0/W12/W24. Saved patient-held-out predictions were evaluated within each repetition before averaging; paired patient resamples were shared across methods. Low IL-22 coverage restricted this matched subset. These were exploratory comparisons, without a confirmatory multiplicity claim (Supplementary Table S15).

OMIX002850 had been examined in earlier project work, possible overlap with Huashan participants remained unresolved, and nominal concentration units did not establish assay equivalence. BSA was absent. Consequently the analysis was a retrospective formula-transfer benchmark, not independent external validation of E-EASI. Published trials and supplement tables were separately checked for compatible inputs; group summaries or relative proteomic signals were not substituted for missing individual absolute concentrations.

1.9 Post hoc comparison with IEC low-activity thresholds

The EASI branch of the IEC disease-control definitions was applied at W0, W4, W12 and W24: low disease activity required EASI $\leq$7 and PP-NRS $\leq$4; very low disease activity required EASI $\leq$3 and PP-NRS $\leq$1.[7] Primary proportions required both scores to be observed at the same nominal visit and used Wilson 95% confidence intervals. Full-population bounds retained known failures when either observed component failed, and assigned unresolved states to failure or success. These descriptive, post hoc analyses neither imputed scores nor tested additional treatment contrasts. They do not reconstruct patient-selected AHEAD priorities or establish continuous on-treatment control or off-treatment remission.

Supplementary tables

N is the endpoint-specific evaluable denominator. Unknown outcomes are excluded from the observed rate and retained in missing-outcome bounds. Adult DLQI requires age at least 18. This table covers the specified visits through W24; exploratory W48 component rates remain in the CSV collection. Exact tests and correction families accompany `endpoint_tests.csv` and `endpoint_weighted_contrasts.csv`.

Table S1: Exact target attainment, observed denominators and Wilson 95% confidence intervals.

Week and target	Initial treatment	Achieved/N	Rate, %	95% CI
W0: Adult DLQI at most 10	abro 100 mg	16/32	50.0	33.6–66.4
W0: Adult DLQI at most 10	abro 200 mg	9/34	26.5	14.6–43.1
W0: Adult DLQI at most 10	dupi	12/47	25.5	15.3–39.5
W0: Adult DLQI at most 4	abro 100 mg	3/32	9.4	3.2–24.2
W0: Adult DLQI at most 4	abro 200 mg	0/34	0.0	0.0–10.2
W0: Adult DLQI at most 4	dupi	4/47	8.5	3.4–19.9
W0: EASI at most 3	abro 100 mg	4/56	7.1	2.8–17.0
W0: EASI at most 3	abro 200 mg	1/42	2.4	0.4–12.3
W0: EASI at most 3	dupi	0/65	0.0	0.0–5.6
W0: EASI at most 5	abro 100 mg	9/56	16.1	8.7–27.8
W0: EASI at most 5	abro 200 mg	3/42	7.1	2.5–19.0
W0: EASI at most 5	dupi	5/65	7.7	3.3–16.8
W0: IGA 0/1	abro 100 mg	0/56	0.0	0.0–6.4
W0: IGA 0/1	abro 200 mg	0/42	0.0	0.0–8.4
W0: IGA 0/1	dupi	0/65	0.0	0.0–5.6
W0: PP-NRS 0/1	abro 100 mg	3/55	5.5	1.9–14.9
W0: PP-NRS 0/1	abro 200 mg	1/42	2.4	0.4–12.3
W0: PP-NRS 0/1	dupi	0/65	0.0	0.0–5.6
W0: PP-NRS at most 4	abro 100 mg	11/55	20.0	11.6–32.4
W0: PP-NRS at most 4	abro 200 mg	7/42	16.7	8.3–30.6
W0: PP-NRS at most 4	dupi	5/65	7.7	3.3–16.8
W2: PP-NRS at most 4	abro 100 mg	34/37	91.9	78.7–97.2
W2: PP-NRS at most 4	abro 200 mg	33/36	91.7	78.2–97.1
W2: PP-NRS at most 4	dupi	15/27	55.6	37.3–72.4
W4: Adult DLQI at most 10	abro 100 mg	14/16	87.5	64.0–96.5
W4: Adult DLQI at most 10	abro 200 mg	13/15	86.7	62.1–96.3
W4: Adult DLQI at most 10	dupi	21/26	80.8	62.1–91.5
W4: Adult DLQI at most 4	abro 100 mg	12/16	75.0	50.5–89.8
W4: Adult DLQI at most 4	abro 200 mg	8/15	53.3	30.1–75.2
W4: Adult DLQI at most 4	dupi	11/26	42.3	25.5–61.1
W4: EASI50	abro 100 mg	26/34	76.5	60.0–87.6
W4: EASI50	abro 200 mg	14/18	77.8	54.8–91.0
W4: EASI50	dupi	14/39	35.9	22.7–51.6
W4: EASI75	abro 100 mg	15/34	44.1	28.9–60.5
W4: EASI75	abro 200 mg	5/18	27.8	12.5–50.9
W4: EASI75	dupi	6/39	15.4	7.2–29.7
W4: EASI90	abro 100 mg	10/34	29.4	16.8–46.2
W4: EASI90	abro 200 mg	3/18	16.7	5.8–39.2
W4: EASI90	dupi	0/39	0.0	0.0–9.0
W4: EASI at most 3	abro 100 mg	21/34	61.8	45.0–76.1
W4: EASI at most 3	abro 200 mg	4/18	22.2	9.0–45.2
W4: EASI at most 3	dupi	4/39	10.3	4.1–23.6
W4: EASI at most 5	abro 100 mg	24/34	70.6	53.8–83.2
W4: EASI at most 5	abro 200 mg	10/18	55.6	33.7–75.4
W4: EASI at most 5	dupi	9/39	23.1	12.6–38.3
W4: IGA 0/1	abro 100 mg	13/34	38.2	23.9–55.0
W4: IGA 0/1	abro 200 mg	2/17	11.8	3.3–34.3
W4: IGA 0/1	dupi	0/39	0.0	0.0–9.0
W4: IGA decrease at least 2	abro 100 mg	18/34	52.9	36.7–68.5
W4: IGA decrease at least 2	abro 200 mg	8/17	47.1	26.2–69.0
W4: IGA decrease at least 2	dupi	8/39	20.5	10.8–35.5
W4: PP-NRS 0/1	abro 100 mg	12/34	35.3	21.5–52.1
W4: PP-NRS 0/1	abro 200 mg	11/18	61.1	38.6–79.7
W4: PP-NRS 0/1	dupi	6/38	15.8	7.4–30.4
W4: PP-NRS decrease at least 4	abro 100 mg	24/29	82.8	65.5–92.4

Week and target	Initial treatment	Achieved/N	Rate, %	95% CI
W4: PP-NRS decrease at least 4	abro 200 mg	14/16	87.5	64.0–96.5
W4: PP-NRS decrease at least 4	dupi	22/36	61.1	44.9–75.2
W4: PP-NRS at most 4	abro 100 mg	33/34	97.1	85.1–99.5
W4: PP-NRS at most 4	abro 200 mg	16/18	88.9	67.2–96.9
W4: PP-NRS at most 4	dupi	26/38	68.4	52.5–80.9
W4: Both domains	abro 100 mg	26/34	76.5	60.0–87.6
W4: Both domains	abro 200 mg	13/17	76.5	52.7–90.4
W4: Both domains	dupi	18/39	46.2	31.6–61.4
W4: Patient domain	abro 100 mg	35/35	100.0	90.1–100.0
W4: Patient domain	abro 200 mg	16/18	88.9	67.2–96.9
W4: Patient domain	dupi	33/37	89.2	75.3–95.7
W4: Physician domain	abro 100 mg	26/34	76.5	60.0–87.6
W4: Physician domain	abro 200 mg	14/17	82.4	59.0–93.8
W4: Physician domain	dupi	18/39	46.2	31.6–61.4
W12: Adult DLQI at most 10	abro 100 mg	22/22	100.0	85.1–100.0
W12: Adult DLQI at most 10	abro 200 mg	24/26	92.3	75.9–97.9
W12: Adult DLQI at most 10	dupi	29/34	85.3	69.9–93.6
W12: Adult DLQI at most 4	abro 100 mg	18/22	81.8	61.5–92.7
W12: Adult DLQI at most 4	abro 200 mg	20/26	76.9	57.9–89.0
W12: Adult DLQI at most 4	dupi	20/34	58.8	42.2–73.6
W12: EASI50	abro 100 mg	38/42	90.5	77.9–96.2
W12: EASI50	abro 200 mg	25/31	80.6	63.7–90.8
W12: EASI50	dupi	38/50	76.0	62.6–85.7
W12: EASI75	abro 100 mg	26/42	61.9	46.8–75.0
W12: EASI75	abro 200 mg	22/31	71.0	53.4–83.9
W12: EASI75	dupi	23/50	46.0	33.0–59.6
W12: EASI90	abro 100 mg	19/42	45.2	31.2–60.1
W12: EASI90	abro 200 mg	7/31	22.6	11.4–39.8
W12: EASI90	dupi	14/50	28.0	17.5–41.7
W12: EASI at most 3	abro 100 mg	29/42	69.0	54.0–80.9
W12: EASI at most 3	abro 200 mg	17/31	54.8	37.8–70.8
W12: EASI at most 3	dupi	23/50	46.0	33.0–59.6
W12: EASI at most 5	abro 100 mg	34/42	81.0	66.7–90.0
W12: EASI at most 5	abro 200 mg	22/31	71.0	53.4–83.9
W12: EASI at most 5	dupi	31/50	62.0	48.2–74.1
W12: IGA 0/1	abro 100 mg	24/42	57.1	42.2–70.9
W12: IGA 0/1	abro 200 mg	9/31	29.0	16.1–46.6
W12: IGA 0/1	dupi	10/50	20.0	11.2–33.0
W12: IGA decrease at least 2	abro 100 mg	26/42	61.9	46.8–75.0
W12: IGA decrease at least 2	abro 200 mg	23/31	74.2	56.8–86.3
W12: IGA decrease at least 2	dupi	25/50	50.0	36.6–63.4
W12: PP-NRS 0/1	abro 100 mg	25/42	59.5	44.5–73.0
W12: PP-NRS 0/1	abro 200 mg	24/31	77.4	60.2–88.6
W12: PP-NRS 0/1	dupi	23/50	46.0	33.0–59.6
W12: PP-NRS decrease at least 4	abro 100 mg	28/36	77.8	61.9–88.3
W12: PP-NRS decrease at least 4	abro 200 mg	27/28	96.4	82.3–99.4
W12: PP-NRS decrease at least 4	dupi	39/48	81.2	68.1–89.8
W12: PP-NRS at most 4	abro 100 mg	37/42	88.1	75.0–94.8
W12: PP-NRS at most 4	abro 200 mg	29/31	93.5	79.3–98.2
W12: PP-NRS at most 4	dupi	43/50	86.0	73.8–93.0
W12: Both domains	abro 100 mg	21/34	61.8	45.0–76.1
W12: Both domains	abro 200 mg	19/30	63.3	45.5–78.1
W12: Both domains	dupi	20/48	41.7	28.8–55.7
W12: Patient domain	abro 100 mg	27/30	90.0	74.4–96.5
W12: Patient domain	abro 200 mg	27/30	90.0	74.4–96.5
W12: Patient domain	dupi	29/41	70.7	55.5–82.4
W12: Physician domain	abro 100 mg	30/42	71.4	56.4–82.8
W12: Physician domain	abro 200 mg	22/31	71.0	53.4–83.9
W12: Physician domain	dupi	26/50	52.0	38.5–65.2
W24: Adult DLQI at most 10	abro 100 mg	12/12	100.0	75.8–100.0
W24: Adult DLQI at most 10	abro 200 mg	16/18	88.9	67.2–96.9
W24: Adult DLQI at most 10	dupi	22/24	91.7	74.2–97.7
W24: Adult DLQI at most 4	abro 100 mg	11/12	91.7	64.6–98.5
W24: Adult DLQI at most 4	abro 200 mg	16/18	88.9	67.2–96.9
W24: Adult DLQI at most 4	dupi	20/24	83.3	64.1–93.3
W24: EASI50	abro 100 mg	29/29	100.0	88.3–100.0

Week and target	Initial treatment	Achieved/N	Rate, %	95% CI
W24: EASI50	abro 200 mg	21/24	87.5	69.0–95.7
W24: EASI50	dupi	35/40	87.5	73.9–94.5
W24: EASI75	abro 100 mg	26/29	89.7	73.6–96.4
W24: EASI75	abro 200 mg	21/24	87.5	69.0–95.7
W24: EASI75	dupi	30/40	75.0	59.8–85.8
W24: EASI90	abro 100 mg	19/29	65.5	47.3–80.1
W24: EASI90	abro 200 mg	14/24	58.3	38.8–75.5
W24: EASI90	dupi	16/40	40.0	26.3–55.4
W24: EASI at most 3	abro 100 mg	23/29	79.3	61.6–90.2
W24: EASI at most 3	abro 200 mg	17/24	70.8	50.8–85.1
W24: EASI at most 3	dupi	22/40	55.0	39.8–69.3
W24: EASI at most 5	abro 100 mg	27/29	93.1	78.0–98.1
W24: EASI at most 5	abro 200 mg	21/24	87.5	69.0–95.7
W24: EASI at most 5	dupi	28/40	70.0	54.6–81.9
W24: IGA 0/1	abro 100 mg	21/29	72.4	54.3–85.3
W24: IGA 0/1	abro 200 mg	15/23	65.2	44.9–81.2
W24: IGA 0/1	dupi	14/40	35.0	22.1–50.5
W24: IGA decrease at least 2	abro 100 mg	24/29	82.8	65.5–92.4
W24: IGA decrease at least 2	abro 200 mg	20/23	87.0	67.9–95.5
W24: IGA decrease at least 2	dupi	27/40	67.5	52.0–79.9
W24: PP-NRS 0/1	abro 100 mg	22/29	75.9	57.9–87.8
W24: PP-NRS 0/1	abro 200 mg	19/24	79.2	59.5–90.8
W24: PP-NRS 0/1	dupi	23/41	56.1	41.0–70.1
W24: PP-NRS decrease at least 4	abro 100 mg	21/25	84.0	65.3–93.6
W24: PP-NRS decrease at least 4	abro 200 mg	21/22	95.5	78.2–99.2
W24: PP-NRS decrease at least 4	dupi	31/38	81.6	66.6–90.8
W24: PP-NRS at most 4	abro 100 mg	29/29	100.0	88.3–100.0
W24: PP-NRS at most 4	abro 200 mg	23/24	95.8	79.8–99.3
W24: PP-NRS at most 4	dupi	37/41	90.2	77.5–96.1
W24: Both domains	abro 100 mg	22/24	91.7	74.2–97.7
W24: Both domains	abro 200 mg	20/23	87.0	67.9–95.5
W24: Both domains	dupi	26/36	72.2	56.0–84.2
W24: Patient domain	abro 100 mg	23/24	95.8	79.8–99.3
W24: Patient domain	abro 200 mg	20/22	90.9	72.2–97.5
W24: Patient domain	dupi	31/35	88.6	74.0–95.5
W24: Physician domain	abro 100 mg	28/29	96.6	82.8–99.4
W24: Physician domain	abro 200 mg	22/24	91.7	74.2–97.7
W24: Physician domain	dupi	32/40	80.0	65.2–89.5

Table S2: Primary performance of the common-model development procedure and matched controls.

	W0	W12	W24
Patients	152	89	75
MAE (95% CI)	3.19 (2.75, 3.70)	1.30 (0.96, 1.71)	1.19 (0.89, 1.53)
BSA-only area MAE	3.12	1.18	1.02
Three-marker-only MAE	7.63	7.57	8.69
Primary minus BSA-only area MAE difference (95% CI)	0.07 (−0.03, 0.17)	0.12 (0.04, 0.20)	0.17 (0.11, 0.24)
RMSE	4.45	2.29	1.87
Mean signed error	−0.59 (−1.27, 0.07)	0.82 (0.43, 1.29)	0.84 (0.50, 1.25)
Spearman correlation	0.918	0.927	0.904
Concordance correlation	0.910	0.900	0.869
R^2	0.819	0.770	0.656
Calibration intercept	2.01 (0.92, 3.28)	0.13 (−0.32, 0.69)	0.07 (−0.20, 0.42)
Calibration slope	0.91 (0.82, 1.00)	0.81 (0.66, 0.94)	0.73 (0.59, 0.82)
Within one point, %	24.0	57.9	59.3
Within two points, %	41.8	84.3	82.0
Severity-band agreement, %	83.2	88.2	92.7
Sensitivity for EASI >7, %	92.5	85.3	100.0
Specificity for EASI ≤7, %	92.1	90.3	91.9

W0/W12 use patient-grouped nested validation; W24 is same-cohort temporal application. Structure and penalty selection are internal to each training fold; earlier marker-panel selection is not repeated. Matched controls use the same patients and outer folds. BSA-only area uses the zero-intercept area structure; three-marker-only uses CCL17, CCL26 and IL-5 without BSA. Positive paired MAE differences mean greater primary-procedure error. Metrics are averaged after scoring each repetition. Intervals are 95% patient-bootstrap intervals conditional on saved predictions; selection and fitting are not repeated. MAE, mean absolute error; RMSE, root mean squared error. Positive signed error indicates overestimation. Severity-band agreement uses four EASI bands for observed and predicted values: ≤ 7, > 7 to ≤ 21, > 21 to ≤ 50, and > 50. Calibration regresses observed EASI on predicted EASI; ideal intercept/slope are 0/1. The single final formula is not independently validated.

Table S3: Coverage of recorded safety documentation by follow-up window.

Initial treatment	Initial N	With note through W24, n/N	No note through W24, n	With note at any follow-up, n/N	Note records through W24, n
Dupilumab	65	27/65	38	28/65	30
Abrocitinib 100 mg	56	20/56	36	20/56	28
Abrocitinib 200 mg	42	19/42	23	19/42	37

Patients are counted once within each documentation window. The through-W24 window uses recorded nominal visits up to W24; the all-follow-up column also includes later recorded visits. Note records can repeat within a patient and are not adverse-event counts. Absence of a note means unknown documentation, not absence of an event. Initial-group N is a cohort denominator, not a verified safety-observation denominator. These counts do not estimate AE or SAE incidence, severity or causality, and do not establish event-time treatment attribution. Source, `safety_documentation_windows.csv`.

Table S4: Robustness to training-only exclusion of reference-inconsistent development visits.

	W0	W12	W24
All original held-out observations			
Patients	152	89	75
Primary MAE (95% CI)	3.19 (2.75, 3.70)	1.30 (0.96, 1.71)	1.19 (0.89, 1.53)
Training-exclusion MAE (95% CI)	3.19 (2.72, 3.70)	1.15 (0.82, 1.57)	1.06 (0.75, 1.40)
Paired ΔMAE (95% CI)	−0.007 (−0.090, 0.078)	−0.149 (−0.228, −0.073)	−0.134 (−0.206, −0.064)
Primary calibration slope	0.913	0.814	0.735
Training-exclusion calibration slope	0.915	0.821	0.733
Paired Δslope (95% CI)	0.002 (−0.008, 0.015)	0.007 (−0.004, 0.028)	−0.002 (−0.015, 0.014)
Strict reference-consistent evaluation subset			
Patients	140	86	71
Primary MAE (95% CI)	3.17 (2.70, 3.71)	1.32 (0.97, 1.75)	1.11 (0.82, 1.42)
Training-exclusion MAE (95% CI)	3.19 (2.72, 3.72)	1.18 (0.83, 1.60)	0.97 (0.70, 1.30)
Paired ΔMAE (95% CI)	0.012 (−0.073, 0.101)	−0.143 (−0.223, −0.066)	−0.131 (−0.203, −0.063)
Primary calibration slope	0.910	0.790	0.765
Training-exclusion calibration slope	0.910	0.798	0.766
Paired Δslope (95% CI)	0.001 (−0.010, 0.014)	0.008 (−0.003, 0.035)	0.001 (−0.013, 0.018)

Eight W0 and two W12 visits with complete four-region scores whose sum differed from recorded total EASI by more than 0.15 points were excluded from development fitting only. Visits with missing regional references were retained. Original patient outer folds, evaluation patients and recorded outcomes were unchanged; preprocessing and inner structure/penalty selection were repeated. The strict evaluation subset requires all four regions and an absolute sum-to-total difference at most 0.01. Paired Δ is training-exclusion minus primary; negative ΔMAE means less error. Calibration regresses observed on predicted EASI, with ideal slope 1. Metrics are scored within each repetition before averaging. Intervals use 2,000 paired patient resamples conditional on the saved predictions (seed 20261002); bootstrap resamples do not repeat fitting or selection. This is robustness to exclusion, not correction of clinical truth, independent validation or a replacement of the primary formula. Matched refitted BSA controls and complete metrics are in the `reference_sensitivity` aggregate directory.

Table S5: Input and clinical-reference specifications, with unresolved source requirements.

Element	Current analysis specification	Evidence boundary
EASI reference	Recorded total EASI, 0–72; observed outcomes only, with no imputed validation truth.	Regional-sum discrepancies are audited; the authoritative field in discrepant records and blinded repeat-assessment reliability are not confirmed.
BSA input	Affected BSA entered from 0 to 100 (10% is entered as 10). It supplies a clinical extent component of the hybrid estimate.	The source does not establish an observer-independent measurement protocol; this is not a blood-only score.
Molecular inputs	CCL17, CCL26 and IL-5. Positive finite integration-export numerics recorded in pg/mL are log2 transformed once.	A positive numeric export is not proof of validated quantification. Transfer to another export or assay requires confirmed scale compatibility.
Specimen and assay	Laboratory reports describe serum, Luminex LXSAHM-13/Luminex-200 and MSD K15067M-1/K15067M-2/SQ120 assays.	Matched specimen handling, technical-repeat precision and cross-platform bridging are not established by the analysis.
Dilution and limits	Reports describe 50-microlitre serum and twofold laboratory dilution. No additional dilution factor is applied to the integration export.	Whether dilution is already corrected in that export, and validated lower/upper quantification limits, require source confirmation. Software calibration boundaries are not validated LLOQ/ULOQ.
Clinical/serum timing	Patient identifier plus nominal W0, W12 or W24 links the clinical and molecular record.	Exact calendar-day separation and a permitted sampling window are unverified; nominal matching does not establish same-day measurement.
Development scope	Age $\geq$12, observed EASI/BSA and all three positive molecular inputs. Fitting and selection use W0/W12 with patient-grouped inner and outer validation.	The marker panel arose from earlier exploration. W24 had previously been inspected and is same-cohort temporal application, not untouched external validation.
Calculator output	The precise frozen coefficients return EASI units clipped to 0–72. Missing, nonfinite or nonpositive molecular inputs and invalid BSA are rejected.	BSA=0 returns zero by the area constraint; it does not prove absence of molecular inflammation. Independent validation of the single final coefficient vector is unavailable.
Recorded IGA	Recorded grades 0–5 are retained. The supplied protocol appendix defines grade 5 as very severe, while its inclusion text says 0–4. No grade is recoded.	Actual implementation across centres/visits and correspondence to a validated IGA system require confirmation. IGA-free sensitivity is in Table S10.
PP-NRS and DLQI	Recorded PP-NRS uses 0–10; the supplied protocol specifies a 24-hour recall window. Adult DLQI uses 0–30 and age $\geq$18. A PP-NRS decrease $\geq$4 requires observed baseline $\geq$4.	Actual recall-window implementation and questionnaire versions remain unconfirmed. Adult DLQI is unknown in adolescent composite rules.
IgE and eosinophils	Analyses retain the recorded numeric values; adjusted models use log(1+x) and back-transformation.	Clinical laboratory units and the meaning of repeated IgE values of 5000 require source confirmation; no censoring threshold is inferred from repetition alone.
Treatment targets	The September investigator framework supplies stage-specific physician and patient thresholds, with the stated three-valued missing-component logic.	The exact approved and prospectively implemented framework version and original voting records are not verified; target attainment is not validation of a new consensus or management effect.

Table S6: Matched input-set performance.

Inputs	W0	W12	W24
MAE in EASI points (95% CI)			
Complete core population; n=152/89/75			
BSA	3.15 (2.71, 3.65)	1.21 (0.87, 1.62)	1.05 (0.75, 1.39)
BSA, fixed area structure	3.12 (2.68, 3.62)	1.18 (0.85, 1.59)	1.02 (0.73, 1.35)
BSA + CCL17	3.16 (2.72, 3.64)	1.25 (0.90, 1.69)	1.15 (0.84, 1.51)
BSA + three markers	3.19 (2.75, 3.70)	1.30 (0.96, 1.71)	1.19 (0.89, 1.53)
BSA + three markers, fixed area structure	3.26 (2.78, 3.76)	1.15 (0.81, 1.56)	1.01 (0.71, 1.33)
CCL17	7.56 (6.61, 8.50)	7.43 (6.56, 8.36)	8.64 (7.64, 9.65)
Three serum markers	7.63 (6.64, 8.70)	7.57 (6.70, 8.51)	8.69 (7.76, 9.66)
Fixed area, without CCL17	3.29 (2.82, 3.80)	1.21 (0.86, 1.63)	1.01 (0.72, 1.35)
Fixed area, without CCL26	3.19 (2.73, 3.69)	1.12 (0.80, 1.54)	1.02 (0.71, 1.35)
Fixed area, without IL-5	3.23 (2.76, 3.71)	1.13 (0.80, 1.54)	0.99 (0.70, 1.31)
Matched current-itch population; n=151/89/75			
BSA	3.16 (2.70, 3.66)	1.21 (0.87, 1.62)	1.05 (0.75, 1.39)
BSA + PP-NRS	3.06 (2.61, 3.54)	1.10 (0.80, 1.47)	0.96 (0.69, 1.27)
BSA + three markers	3.20 (2.72, 3.71)	1.29 (0.95, 1.71)	1.19 (0.89, 1.53)
BSA + three markers + PP-NRS	3.08 (2.62, 3.57)	1.17 (0.86, 1.57)	1.04 (0.76, 1.38)
PP-NRS + three markers	7.14 (6.23, 8.15)	4.54 (3.87, 5.23)	4.05 (3.42, 4.69)
Three serum markers	7.67 (6.65, 8.77)	7.48 (6.60, 8.43)	8.56 (7.63, 9.54)
Matched adult itch and DLQI population; n=105/58/44			
BSA	2.72 (2.28, 3.19)	1.23 (0.82, 1.83)	1.12 (0.70, 1.61)
BSA + PP-NRS + DLQI	2.73 (2.30, 3.22)	1.13 (0.75, 1.65)	1.00 (0.61, 1.43)
BSA + three markers	2.82 (2.35, 3.33)	1.34 (0.92, 1.92)	1.22 (0.79, 1.71)
BSA + three markers + DLQI	2.86 (2.41, 3.35)	1.12 (0.74, 1.65)	0.96 (0.57, 1.42)
BSA + three markers + PP-NRS	2.80 (2.33, 3.30)	1.17 (0.77, 1.75)	1.01 (0.59, 1.47)
BSA + three markers + PP-NRS + DLQI	2.83 (2.39, 3.33)	1.13 (0.75, 1.67)	0.92 (0.54, 1.36)
Spearman / concordance correlation			
Complete core population; n=152/89/75			
BSA	0.923 / 0.913	0.944 / 0.906	0.937 / 0.883
BSA, fixed area structure	0.924 / 0.915	0.943 / 0.909	0.937 / 0.887
BSA + CCL17	0.915 / 0.913	0.936 / 0.897	0.924 / 0.866
BSA + three markers	0.918 / 0.910	0.927 / 0.900	0.904 / 0.869
BSA + three markers, fixed area structure	0.921 / 0.911	0.939 / 0.908	0.934 / 0.887
CCL17	0.345 / 0.356	-0.102 / 0.085	0.076 / 0.060
Three serum markers	0.384 / 0.329	0.071 / 0.103	0.114 / 0.051
Fixed area, without CCL17	0.920 / 0.908	0.940 / 0.906	0.934 / 0.886
Fixed area, without CCL26	0.922 / 0.914	0.941 / 0.909	0.938 / 0.885
Fixed area, without IL-5	0.922 / 0.912	0.941 / 0.911	0.934 / 0.891
Matched current-itch population; n=151/89/75			
BSA	0.922 / 0.912	0.944 / 0.906	0.937 / 0.883
BSA + PP-NRS	0.916 / 0.913	0.953 / 0.920	0.937 / 0.902
BSA + three markers	0.917 / 0.909	0.928 / 0.900	0.904 / 0.869
BSA + three markers + PP-NRS	0.912 / 0.912	0.944 / 0.911	0.935 / 0.887
PP-NRS + three markers	0.431 / 0.426	0.297 / 0.312	0.337 / 0.377
Three serum markers	0.378 / 0.335	0.049 / 0.101	0.101 / 0.055
Matched adult itch and DLQI population; n=105/58/44			
BSA	0.925 / 0.938	0.933 / 0.844	0.966 / 0.902
BSA + PP-NRS + DLQI	0.924 / 0.934	0.934 / 0.866	0.959 / 0.920
BSA + three markers	0.920 / 0.932	0.929 / 0.837	0.957 / 0.896
BSA + three markers + DLQI	0.918 / 0.929	0.932 / 0.856	0.970 / 0.916
BSA + three markers + PP-NRS	0.914 / 0.930	0.935 / 0.851	0.969 / 0.912
BSA + three markers + PP-NRS + DLQI	0.915 / 0.930	0.937 / 0.862	0.969 / 0.922

All models within a population use identical observations and patient folds. Three markers are CCL17, CCL26 and IL-5. BSA and questionnaires are from the current visit; DLQI requires age at least 18. W0/W12 training precedes previously inspected W24 temporal application. Core and adult populations must not be compared as though their participants were identical. Fixed-area rows hold the model structure constant; other BSA models select among linear, quadratic and area structures inside training folds. Intervals use 2,000 paired patient resamples conditional on saved predictions; all comparisons are exploratory.

Table S7: Paired differences in mean absolute error.

Comparison	W0	W12	W24
Three markers versus CCL17	+0.073 (-0.228, +0.428)	+0.140 (-0.212, +0.509)	+0.049 (-0.356, +0.451)
BSA + three markers versus BSA + CCL17	+0.032 (-0.062, +0.135)	+0.050 (-0.030, +0.131)	+0.042 (-0.046, +0.117)
BSA + three markers versus BSA	+0.041 (-0.060, +0.149)	+0.089 (+0.006, +0.178)	+0.140 (+0.073, +0.207)
Same area structure: add three markers	+0.133 (-0.001, +0.280)	-0.031 (-0.086, +0.032)	-0.015 (-0.044, +0.014)
BSA + three markers: add PP-NRS	-0.125 (-0.277, +0.026)	-0.127 (-0.236, -0.011)	-0.145 (-0.246, -0.047)
BSA + PP-NRS: add three markers	+0.020 (-0.078, +0.123)	+0.064 (-0.012, +0.147)	+0.082 (+0.014, +0.162)
Adult BSA + three markers + PP-NRS: add DLQI	+0.035 (-0.096, +0.179)	-0.041 (-0.132, +0.045)	-0.082 (-0.172, -0.000)
Adult BSA + PP-NRS + DLQI: add three markers	+0.107 (+0.035, +0.186)	+0.006 (-0.035, +0.047)	-0.072 (-0.124, -0.021)

Differences are the first or augmented model minus its stated reference; negative values indicate less error. Confidence intervals use the same patient resample for both models. These exploratory intervals are not adjusted for the many input comparisons, and do not establish clinical importance. The adult rows use 105/58/44 patients; itch rows use 151/89/75, and core rows use 152/89/75. In particular, a sub-point adult W24 error cannot be compared directly with a larger full-cohort error.

Table S8: IgE observation patterns and missing-data sensitivity.

Initial group / analysis	W0–W12	W0–W24	W12–W24
Observed all-centre paired ratios of IgE+1 (95% CI); paired n			
dupi	0.51 (0.39, 0.67); 29	0.34 (0.20, 0.57); 14	0.65 (0.57, 0.75); 12
abro 100 mg	0.56 (0.40, 0.79); 19	0.66 (0.40, 1.11); 12	1.06 (0.88, 1.27); 11
abro 200 mg	0.45 (0.18, 1.14); 20	0.82 (0.50, 1.36); 13	1.16 (0.73, 1.82); 11
Posterior log-normal imputation: CAMS, MAR			
dupi	0.55 (0.35, 0.87)	0.37 (0.22, 0.62)	0.67 (0.48, 0.93)
abro 100 mg	0.56 (0.30, 1.07)	0.63 (0.31, 1.27)	1.12 (0.74, 1.69)
abro 200 mg	0.56 (0.27, 1.14)	0.69 (0.31, 1.52)	1.24 (0.80, 1.91)
Within-treatment donor imputation: CAMS, MAR			
dupi	0.58 (0.41, 0.84)	0.43 (0.29, 0.64)	0.73 (0.53, 1.02)
abro 100 mg	0.50 (0.28, 0.90)	0.68 (0.39, 1.19)	1.37 (0.78, 2.39)
abro 200 mg	0.78 (0.34, 1.75)	1.92 (0.88, 4.18)	2.48 (1.14, 5.41)

CAMS W0–W24 sensitivity to missing-value assumptions

Initial group / method	Half	MAR	Double
dupi; Log-normal	0.23 (0.13, 0.38)	0.37 (0.22, 0.62)	0.60 (0.35, 1.02)
abro 100 mg; Log-normal	0.39 (0.19, 0.78)	0.63 (0.31, 1.27)	1.03 (0.51, 2.09)
abro 200 mg; Log-normal	0.43 (0.19, 0.96)	0.69 (0.31, 1.52)	1.10 (0.50, 2.45)
dupi; Donor	0.26 (0.18, 0.39)	0.43 (0.29, 0.64)	0.70 (0.46, 1.06)
abro 100 mg; Donor	0.42 (0.24, 0.74)	0.68 (0.39, 1.19)	1.12 (0.63, 1.97)
abro 200 mg; Donor	1.20 (0.56, 2.56)	1.92 (0.88, 4.18)	3.08 (1.37, 6.93)

Ratios exponentiate mean changes in log(IgE+1); they are not arithmetic mean ratios. All-centre observed pairs and CAMS imputation have different target populations. The second block varies missing-value assumptions for W0–W24. MAR assumes that included observed information explains missingness; half/double multiply only missing follow-up (IgE+1) by 0.5/2, retaining the physical lower bound, and illustrate departures from MAR. Imputation targets 130 CAMS patients, with 400 retained completed datasets; these are not 400 independent patient cohorts. Complete all-centre sensitivity, centre-specific coverage, observation-pattern summaries, donor support and numerical diagnostics remain in the source tables. No IgE was observed at W4 and none was created. The value 5000 is retained as recorded; its assay-censoring status is unverified.

Table S9: Laboratory coverage and observed paired eosinophil changes

Measure	Initial group	W0	W4	W12	W24
IgE	dupi	62/65	0/65	30/65	15/65
IgE	abro 100 mg	41/56	0/56	23/56	14/56
IgE	abro 200 mg	37/42	0/42	22/42	14/42
Eosinophils	dupi	34/65	0/65	20/65	17/65
Eosinophils	abro 100 mg	51/56	28/56	39/56	24/56
Eosinophils	abro 200 mg	38/42	16/42	29/42	24/42

Observed/initial patients. Zero denotes no recorded measurement, not a laboratory value of zero. Patients may differ between visits. All values remain in their recorded units. No W4 IgE or dupi eosinophil point was imputed.

Interval	Initial group	Pairs	Mean eosinophil change (95% CI)
W0–W12	dupi	15	−0.124 (−0.304, 0.056)
W0–W12	abro 100 mg	36	−0.158 (−0.289, −0.028)
W0–W12	abro 200 mg	26	−0.308 (−0.439, −0.178)
W0–W24	dupi	10	−0.158 (−0.405, 0.089)
W0–W24	abro 100 mg	23	−0.089 (−0.208, 0.030)
W0–W24	abro 200 mg	22	−0.263 (−0.442, −0.084)
W12–W24	dupi	10	−0.117 (−0.201, −0.033)
W12–W24	abro 100 mg	22	0.008 (−0.055, 0.071)
W12–W24	abro 200 mg	20	0.086 (−0.000, 0.173)

Later minus earlier measurements in all available pairs, without requiring the third visit. Descriptive t-based intervals are not multiplicity-adjusted treatment-comparison tests. Baseline-adjusted group contrasts and centre-specific coverage remain in the aggregate sources. No follow-up eosinophil measurements were recorded for Huashan dupi patients; an overall observed change cannot represent that unobserved cell.

Table S10: Combined-target sensitivity without an IGA component

Week	Initial group	Paired N	Recorded	No IGA	Changed
W4	dupi	39	18/39	14/39	4
W4	abro 100 mg	34	26/34	26/34	0
W4	abro 200 mg	17	13/17	13/17	0
W12	dupi	48	20/48	18/48	2
W12	abro 100 mg	34	21/34	17/34	4
W12	abro 200 mg	30	19/30	19/30	0
W24	dupi	36	26/36	26/36	0
W24	abro 100 mg	24	22/24	22/24	0
W24	abro 200 mg	23	20/23	20/23	0

Both definitions are evaluated in the same patients within each row. Recorded denotes the reconstructed framework, not a verified consensus endpoint. No IGA removes only the IGA alternative from the physician domain, retaining EASI50 at W4, EASI75 at W12 and EASI75 or EASI$\leq$ 5 at W24; the patient domain is unchanged. This is a different, exploratory target definition, not a recoding of IGA.

Week	Initial group	Recorded, all evaluable	No IGA, all evaluable
W4	dupi	18/39	14/39
W4	abro 100 mg	26/34	26/34
W4	abro 200 mg	13/17	13/18
W12	dupi	20/48	18/48
W12	abro 100 mg	21/34	17/34
W12	abro 200 mg	19/30	19/30
W24	dupi	26/36	26/36
W24	abro 100 mg	22/24	22/24
W24	abro 200 mg	20/23	20/23

The second block uses each definition's own evaluable population; a change in denominator is not a change in the same patient's response. Unknown components follow three-valued AND/OR rules and are never filled with clinical scores. Full transition counts, physician-domain results and exclusion of the 22 patients with recorded baseline IGA5 are supplied in the source tables. The protocol inclusion text specifies 0–4 but its IGA appendix defines grades 0–5, including grade 5 as very severe. All original grades are therefore retained; actual implementation requires confirmation.

Table S11: Response classification by validation repetition

Visit / target	Repeat	Sensitivity	Specificity	Agreement
		64/70	14/15	78/85
W12 / EASI50	1	91.4 (82.5, 96.0)	93.3 (70.2, 98.8)	91.8 (84.0, 96.0)
		66/70	14/15	80/85
W12 / EASI50	2	94.3 (86.2, 97.8)	93.3 (70.2, 98.8)	94.1 (87.0, 97.5)
		36/49	36/36	72/85
W12 / EASI75	1	73.5 (59.7, 83.8)	100.0 (90.4, 100.0)	84.7 (75.6, 90.8)
		39/49	35/36	74/85
W12 / EASI75	2	79.6 (66.4, 88.5)	97.2 (85.8, 99.5)	87.1 (78.3, 92.6)
		17/27	57/58	74/85
W12 / EASI90	1	63.0 (44.2, 78.5)	98.3 (90.9, 99.7)	87.1 (78.3, 92.6)
		17/27	56/58	73/85
W12 / EASI90	2	63.0 (44.2, 78.5)	96.6 (88.3, 99.0)	85.9 (76.9, 91.7)
		57/60	7/7	64/67
W24 / EASI50	1	95.0 (86.3, 98.3)	100.0 (64.6, 100.0)	95.5 (87.6, 98.5)
		57/60	7/7	64/67
W24 / EASI50	2	95.0 (86.3, 98.3)	100.0 (64.6, 100.0)	95.5 (87.6, 98.5)
		46/54	13/13	59/67
W24 / EASI75	1	85.2 (73.4, 92.3)	100.0 (77.2, 100.0)	88.1 (78.2, 93.8)
		47/54	12/13	59/67
W24 / EASI75	2	87.0 (75.6, 93.6)	92.3 (66.7, 98.6)	88.1 (78.2, 93.8)
		18/35	30/32	48/67
W24 / EASI90	1	51.4 (35.6, 67.0)	93.8 (79.9, 98.3)	71.6 (59.9, 81.0)
		19/35	30/32	49/67
W24 / EASI90	2	54.3 (38.2, 69.5)	93.8 (79.9, 98.3)	73.1 (61.5, 82.3)

Each cell gives correctly classified patients/eligible patients and percentage (Wilson 95% interval). The repetitions contain the same patients and are not pooled as independent observations. The main figure averages repetition-specific percentages. Its conservative bounds average limits of two-sided 97.5% Clopper–Pearson intervals, using Bonferroni across the two repetitions within each target, visit and metric. These bounds assume independent Bernoulli patient outcomes within each repetition, permit dependence between repetitions, and condition on saved predictions. They do not propagate shared training fits or model-development uncertainty and do not provide simultaneous coverage across all targets and metrics. W24 remains same-cohort temporal application.

Table S12: Available pretreatment concentrations of all 23 serum markers.

Marker	Dupilumab	Abrocitinib 100 mg	Abrocitinib 200 mg
CCL11	99.19 (88.64–114.12)	97.38 (87.33–116.27)	103.39 (95.46–126.13)
	n=59; outside range=0	n=54; outside range=0	n=41; outside range=0
CCL17	798.67 (462.16–1369.00)	673.25 (426.61–1005.86)	1232.00 (713.64–2435.00)
	n=59; outside range=0	n=54; outside range=0	n=41; outside range=0
CCL18	12103.00 (10028.00–14058.00)	11880.00 (10310.50–12837.50)	13069.00 (11234.50–14926.25)
	n=37; outside range=22	n=31; outside range=23	n=18; outside range=23
CCL26	9.85 (8.52–11.53)	8.73 (7.53–10.41)	10.41 (8.73–12.02)
	n=59; outside range=0	n=51; outside range=3	n=41; outside range=0
IFN-beta	0.81 (0.50–1.12)	0.69 (0.34–1.12)	0.51 (0.35–0.93)
	n=52; outside range=0	n=42; outside range=0	n=38; outside range=0
IFN-gamma	16.76 (11.29–25.79)	18.19 (11.04–29.20)	19.98 (13.81–29.10)
	n=59; outside range=0	n=54; outside range=0	n=41; outside range=0
IL-10	0.51 (0.43–0.66)	0.50 (0.46–0.64)	0.51 (0.45–0.69)
	n=59; outside range=0	n=54; outside range=0	n=41; outside range=0
IL-12p70	0.11 (0.07–0.22)	0.14 (0.07–0.23)	0.16 (0.09–0.25)
	n=59; outside range=0	n=51; outside range=0	n=41; outside range=0
IL-13	0.82 (0.45–1.44)	0.78 (0.32–1.18)	1.36 (0.79–2.21)
	n=32; outside range=0	n=20; outside range=0	n=10; outside range=0
IL-17A	5.61 (5.48–6.44)	6.11 (4.57–7.29)	7.75 (6.11–9.48)
	n=58; outside range=1	n=49; outside range=5	n=41; outside range=0
IL-18	366.21 (274.37–471.58)	381.15 (297.61–524.83)	459.69 (393.60–634.40)
	n=59; outside range=0	n=53; outside range=1	n=41; outside range=0
IL-21	0.99 (0.54–1.87)	0.96 (0.58–2.73)	1.08 (0.74–2.02)
	n=51; outside range=0	n=33; outside range=0	n=27; outside range=0
IL-22	1.12 (0.31–2.03)	0.62 (0.35–1.54)	2.34 (1.02–3.43)
	n=23; outside range=0	n=11; outside range=0	n=26; outside range=0
IL-31	6.03 (3.42–10.59)	4.61 (3.13–9.76)	7.44 (5.78–10.72)
	n=31; outside range=0	n=28; outside range=0	n=22; outside range=0
IL-33	15.34 (13.58–17.11)	16.98 (15.27–18.91)	17.15 (15.27–18.09)
	n=59; outside range=0	n=54; outside range=0	n=41; outside range=0
IL-36beta	12.62 (12.46–13.70)	12.54 (11.60–13.26)	13.21 (11.87–13.89)
	n=59; outside range=0	n=54; outside range=0	n=41; outside range=0
IL-4	83.26 (79.09–90.48)	91.26 (83.39–95.48)	91.52 (87.49–99.37)
	n=59; outside range=0	n=54; outside range=0	n=41; outside range=0
IL-5	1.07 (0.57–2.10)	0.81 (0.60–1.56)	0.91 (0.59–1.73)
	n=59; outside range=0	n=54; outside range=0	n=41; outside range=0
IL-6	1.06 (0.63–1.65)	0.89 (0.61–1.43)	0.97 (0.68–1.60)
	n=59; outside range=0	n=54; outside range=0	n=41; outside range=0
IL2RA	499.34 (415.06–663.43)	562.21 (418.86–658.90)	593.76 (478.31–760.60)
	n=59; outside range=0	n=54; outside range=0	n=41; outside range=0
Periostin	82006.00 (66842.50–94006.50)	70030.50 (57474.50–93283.00)	87900.00 (65214.00–124773.00)
	n=59; outside range=0	n=54; outside range=0	n=41; outside range=0
TNF-alpha	4.61 (3.87–5.56)	3.84 (3.15–4.61)	3.86 (3.38–4.88)
	n=59; outside range=0	n=54; outside range=0	n=41; outside range=0
TSLP	6.47 (5.92–7.08)	7.05 (6.40–8.04)	7.17 (6.73–8.26)
	n=59; outside range=0	n=54; outside range=0	n=41; outside range=0

Values are median (Q1–Q3) in recorded pg/mL, calculated on individual original-scale concentrations. n counts retained positive numeric results; outside range counts reported below- or above-quantification results excluded from these summaries. Other missing or nonnumeric entries are not included in either count. The table therefore describes the retained numeric sample, not an imputed full-group distribution. Positive numeric export values do not establish analytical validity or verified clinical quantification limits. Quartiles use linear interpolation. Complete detection-status counts are in `baseline_marker_summary.csv`.

Table S13: Published treatment-target frameworks and the local operational schedule.

Framework / timing	Decision rule	Clinical and patient targets	Use in this cohort
International T2T (2021) 3 and 6 months Source	PtGA target plus ≥1 disease-domain target PtGA decrease ≥1 then ≤2; treatment choices also consider safety and tolerability	EASI-50 at 3 months; EASI-75 or EASI ≤7 at 6 months Itch decrease ≥3 at 3 months; PP-NRS ≤4 at 6 months. DLQI decrease ≥4 then ≤5; POEM decrease ≥4 then ≤7	Can reproduce individual EASI/itch/DLQI components, but full algorithm requires verified PtGA
Chinese systemic-treatment guidance (2022) Weeks 2, 4, 8, 12; 6 and 12 months Source	Several clinical and patient-reported domains; ADCT added for control ADCT <7 from Week 12	EASI-50 early; EASI-75 or EASI ≤7 at later assessments Week 2: PP-NRS decrease ≥4, not PP-NRS ≤4. Later PP-NRS ≤4; DLQI ≤5 and POEM ≤7 by Week 12	Only measured domains can be reproduced; missing ADCT/POEM prevents full framework emulation
Canadian T2T (2023) 12–16 weeks; 6–8 months; 1 year Source	≥1 clinician-reported AND ≥1 patient-reported target Topical care continues; unmet targets prompt treatment review	EASI-50/PGA ≤2; EASI-75/PGA ≤2; EASI ≤7/EASI-90/PGA 0/1 Early improvement targets progress to absolute itch ≤4, DLQI ≤5 or POEM ≤7 at 1 year	Only the early window overlaps directly; Week 24 is not the 1-year assessment
AHEAD / MDA (2024) Review within 3–6 months Source	Patient chooses priority features; clinician selects ≥1 clinical measure; optimal agreed targets define MDA Patient priorities determine selected targets; this is not one universal fixed composite	Examples: EASI-90 or EASI ≤3; alternatively IGA 0/1 plus BSA ≤2% If selected: itch ≤1, DLQI 0/1, POEM ≤2, sleep ≤1, pain ≤1; mental health has separate targets	Can report a prespecified EASI-and-itch operationalization, not full individualized AHEAD attainment without chosen-priority records
STRIVE AD (2026) Immediate <4 weeks; short term 3–6 months; long term ≥1 year Source	Shared targets; clear/almost-clear skin and minimal itch Long-term goals include sustained control, fewer flares and reduced topical steroid use	Aspire to EASI-90 or EASI ≤3 at 4–6 months Pruritus NRS ≤1 and DLQI ≤3 at 4–6 months	Endpoint components measurable; durability and patient priorities require additional longitudinal records
IEC disease-control states (2026) Single-visit LDA/vLDA; ≥6 months on-drug control; ≥12 months off-drug remission Source	Clinical signs AND itch A single visit meeting clear-skin thresholds does not establish complete control or remission	EASI branch: LDA ≤7; vLDA ≤3; complete-control/remission threshold EASI =0 PP-NRS ≤4 for LDA; ≤1 for vLDA and control/remission	Use EASI branch plus itch; sparse Week 0/12/24 visits cannot establish continuous 6-month control or 12-month off-drug remission
Local Chinese project framework Weeks 2, 4, 12, 24 in this analysis	Week 2 itch only; then ≥1 clinical-sign AND ≥1 patient-experience target Months 9/12 EASI-90 or EASI ≤3 are planned later targets, not evaluated cohort follow-up	Week 4 EASI-50/IGA decrease ≥2; Week 12 EASI-75/IGA 0/1; Week 24 adds EASI ≤5 Week 2/4 itch ≤4; Week 4 DLQI ≤10 alternative; Week 12/24 itch ≤1 or DLQI ≤4	Observed-visit implementation; quorum and vote-by-item evidence still require original ballots

MDA, minimal disease activity; LDA/vLDA, low/very low disease activity; PtGA, patient global assessment; POEM, Patient-Oriented Eczema Measure. These frameworks are not interchangeable. AHEAD MDA depends on patient-selected priorities. IEC complete control and remission require duration criteria in addition to a low score. The 2022 Chinese guidance thresholds were checked against the 2026 publication by the guidance authors because the original Chinese full text was not available. The local source framework also plans months 9/12: EASI-90, IGA 0/1 or EASI ≤3 in the clinical domain, and PP-NRS 0/1 or DLQI ≤4 in the patient domain. Those later targets were not assessed in the 24-week cohort. Source-specific verification details are in `framework_comparison.csv`.

Table S14: Molecular-only model extensions evaluated in the same patients and folds.

Model	Week	n	MAE (95% CI)	Spearman	CCC
Three markers: ridge	0	152	7.63 (6.64–8.70)	0.384	0.329
Three markers: ridge	12	89	7.57 (6.70–8.51)	0.071	0.103
Three markers: ridge	24	75	8.69 (7.76–9.66)	0.114	0.051
Three markers: nested candidate selection	0	152	7.50 (6.54–8.51)	0.378	0.361
Three markers: nested candidate selection	12	89	7.21 (6.35–8.13)	0.104	0.131
Three markers: nested candidate selection	24	75	7.80 (6.89–8.79)	0.061	0.060
19 markers: ridge	0	152	7.53 (6.60–8.49)	0.357	0.401
19 markers: ridge	12	89	7.28 (6.43–8.19)	0.066	0.093
19 markers: ridge	24	75	8.39 (7.47–9.35)	0.239	0.097
19 markers: paired nonlinear	0	152	7.51 (6.66–8.40)	0.338	0.420
19 markers: paired nonlinear	12	89	7.08 (6.29–7.97)	0.097	0.122
19 markers: paired nonlinear	24	75	8.31 (7.32–9.29)	0.176	0.091
19 markers: IGA auxiliary	0	152	7.53 (6.61–8.48)	0.353	0.403
19 markers: IGA auxiliary	12	89	7.24 (6.39–8.17)	0.078	0.099
19 markers: IGA auxiliary	24	75	8.39 (7.49–9.36)	0.243	0.099
19 markers: inner selection	0	152	7.41 (6.57–8.29)	0.351	0.424
19 markers: inner selection	12	89	7.26 (6.48–8.14)	0.088	0.116
19 markers: inner selection	24	75	8.49 (7.54–9.46)	0.177	0.084

All rows use the same legacy core evaluation set, complete for BSA and the three-marker panel, with 152/89/75 patients at W0/W12/W24. Other serum values are imputed within training folds. Only molecular concentrations are inputs at prediction. The three-marker set is CCL17, CCL26 and IL-5; the 19-marker set follows the prespecified assay-quality screen. Training uses W0/W12, with the existing two repeats of five patient-level outer folds and three inner folds. Imputation, scaling and tuning are training-fold operations. The candidate set permits ridge, paired nonlinear constraints (including a zero-constraint setting) and IGA-assisted reduced-rank estimation; the inner criterion selects the candidate without using held-out scores. IGA is an auxiliary training outcome, not a prediction input. W24 is previously inspected same-cohort temporal application. Patient-bootstrap intervals condition on the fitted predictions and do not include all model-selection uncertainty. The extensions are exploratory after prior inspection of this cohort; they do not replace the frozen E-EASI equation. Complete RMSE, bias, calibration, severity strata, paired-change results, expanded-training populations and paired comparisons are in the aggregate CSV tables.

Table S15: Unchanged serum equations applied to matched local and OMIX002850 samples.

Dataset / visit	Equation	n	MAE (95% CI)	Spearman	CCC
OMIX W0	Local three-marker serum	125	11.51 (9.87–13.28)	0.127	0.075
OMIX W0	Published p-EASI	125	11.27 (9.75–12.90)	0.140	0.128
OMIX W16	Local three-marker serum	60	6.92 (6.20–7.69)	0.557	0.332
OMIX W16	Published p-EASI	60	3.89 (2.96–4.90)	0.364	0.173
Local W0	Local three-marker serum	60	9.16 (7.42–11.11)	0.455	0.334
Local W0	Published p-EASI	60	9.25 (7.41–11.24)	0.462	0.394
Local W12	Local three-marker serum	12	10.42 (7.98–13.01)	0.168	0.169
Local W12	Published p-EASI	12	3.84 (2.33–5.71)	0.014	0.122
Local W24	Local three-marker serum	13	7.80 (4.36–11.57)	0.154	0.062
Local W24	Published p-EASI	13	4.93 (3.48–6.51)	-0.082	-0.073

The serum-only model is a separately fitted CCL17/CCL26/IL-5 equation, not the hybrid E-EASI equation with BSA deleted. Local evaluations use held-out predictions and identical eligible records; published p-EASI is applied unchanged. OMIX estimates use the frozen serum equation without refitting; 125 pretreatment and 60 week-16 AD samples are available. Healthy controls are excluded. Both calculations are clipped to 0–72 here; raw outputs and the rounded 2020 p-EASI version are retained in source tables. OMIX lacks BSA, was previously inspected, and may share patients with the local source centre; participant nonoverlap and assay equivalence are unverified. This is a transport benchmark, not untouched independent validation of E-EASI. The local IL2RA assay is an unbridged proxy for published sIL-2R. The three-marker formula did not establish superiority over p-EASI.

Table S16: Single-visit low and very low disease activity using the IEC EASI-and-itch branch.

Initial treatment	Week	Joint n / full N	LDA	vLDA
Dupilumab	0	65 / 65	0/65: 0.0% (0.0–5.6) Full-group bounds: 0.0–0.0%	0/65: 0.0% (0.0–5.6) Full-group bounds: 0.0–0.0%
Dupilumab	4	38 / 65	12/38: 31.6% (19.1–47.5) Full-group bounds: 18.5–58.5%	1/38: 2.6% (0.5–13.5) Full-group bounds: 1.5–41.5%
Dupilumab	12	50 / 65	32/50: 64.0% (50.1–75.9) Full-group bounds: 49.2–72.3%	15/50: 30.0% (19.1–43.8) Full-group bounds: 23.1–46.2%
Dupilumab	24	40 / 65	30/40: 75.0% (59.8–85.8) Full-group bounds: 46.2–84.6%	18/40: 45.0% (30.7–60.2) Full-group bounds: 27.7–66.2%
Abro 100 mg	0	55 / 56	3/55: 5.5% (1.9–14.9) Full-group bounds: 5.4–5.4%	0/55: 0.0% (0.0–6.5) Full-group bounds: 0.0–0.0%
Abro 100 mg	4	34 / 56	25/34: 73.5% (56.9–85.4) Full-group bounds: 44.6–83.9%	9/34: 26.5% (14.6–43.1) Full-group bounds: 16.1–55.4%
Abro 100 mg	12	42 / 56	35/42: 83.3% (69.4–91.7) Full-group bounds: 62.5–87.5%	20/42: 47.6% (33.4–62.3) Full-group bounds: 35.7–60.7%
Abro 100 mg	24	29 / 56	29/29: 100.0% (88.3–100.0) Full-group bounds: 51.8–100.0%	20/29: 69.0% (50.8–82.7) Full-group bounds: 35.7–83.9%
Abro 200 mg	0	42 / 42	4/42: 9.5% (3.8–22.1) Full-group bounds: 9.5–9.5%	1/42: 2.4% (0.4–12.3) Full-group bounds: 2.4–2.4%
Abro 200 mg	4	18 / 42	11/18: 61.1% (38.6–79.7) Full-group bounds: 26.2–83.3%	2/18: 11.1% (3.1–32.8) Full-group bounds: 4.8–61.9%
Abro 200 mg	12	31 / 42	24/31: 77.4% (60.2–88.6) Full-group bounds: 57.1–83.3%	15/31: 48.4% (32.0–65.2) Full-group bounds: 35.7–61.9%
Abro 200 mg	24	24 / 42	22/24: 91.7% (74.2–97.7) Full-group bounds: 52.4–95.2%	16/24: 66.7% (46.7–82.0) Full-group bounds: 38.1–81.0%
All treatments	0	162 / 163	7/162: 4.3% (2.1–8.6) Full-group bounds: 4.3–4.3%	1/162: 0.6% (0.1–3.4) Full-group bounds: 0.6–0.6%
All treatments	4	90 / 163	48/90: 53.3% (43.1–63.3) Full-group bounds: 29.4–73.6%	12/90: 13.3% (7.8–21.9) Full-group bounds: 7.4–51.5%
All treatments	12	123 / 163	91/123: 74.0% (65.6–80.9) Full-group bounds: 55.8–80.4%	50/123: 40.7% (32.4–49.5) Full-group bounds: 30.7–55.2%
All treatments	24	93 / 163	81/93: 87.1% (78.8–92.5) Full-group bounds: 49.7–92.6%	54/93: 58.1% (47.9–67.6) Full-group bounds: 33.1–76.1%

LDA requires EASI $\leq$7 AND PP-NRS $\leq$4; vLDA requires EASI $\leq$3 AND PP-NRS $\leq$1. Percentages and Wilson 95% confidence intervals use patients with both observed scores at the same nominal visit. Joint n is that observed denominator; full N is the initial treatment population. Full-group bounds assign every unresolved AND state to failure or success, while any observed failing component remains a failure. These bounds are missing-outcome scenarios, not confidence intervals. They can differ from the observed-joint rate because a partially measured patient can already be known to fail. The analysis is post hoc, with no imputed clinical scores or new between-treatment significance tests. It evaluates a single-visit state, not individualized AHEAD MDA, six-month continuous control, remission or a causal treatment effect. All components and unknown counts are in `joint_low_activity_rates.csv` and `joint_low_activity_rates_overall.csv`.

Independent auxiliary target models

Auxiliary clinical targets. We fitted separate models for PP-NRS, adult DLQI and the recorded six-category IGA using the same restricted inputs as the EASI model: BSA, CCL17, CCL26 and IL-5. Continuous targets used ridge regression on the original or log-transformed scale, selected by patient-level inner validation and bounded to the scale range. Recorded IGA used a proportional-odds ordinal logistic model, selected by normalized ranked probability score (RPS). All fits used observed baseline and week-12 outcomes. Week 24 was excluded from model selection. DLQI analyses were restricted to participants aged at least 18 years. These models estimated separate targets; the EASI coefficients were not reused as their prediction formulas.

The common input panel did not yield comparably accurate estimates of itch or quality of life. PP-NRS MAE was 2.35, 2.78 and 2.53 points at baseline, week 12 and week 24, respectively (151, 89 and 75 participants). Adult DLQI MAE was 6.25, 5.90 and 6.28 points (105, 58 and 44 participants). For recorded IGA, normalized RPS was 0.071, 0.099 and 0.124, with exact-category agreement of 54.3%, 35.4% and 28.0% (152, 89 and 75 participants). Adding the three molecular measurements to BSA did not produce a consistent improvement for PP-NRS or DLQI, and ordinal performance was poorer than the BSA-only reference at weeks 12 and 24. These findings limit the interpretation of the common panel to the outcomes for which it was informative. They do not establish that every independently selected panel would fail. The exact identity of the recorded 0–5 IGA instrument requires confirmation, so these ordinal results are reported as exploratory analyses of recorded grades.

Transport of published equations

Published p-EASI coefficients were applied unchanged to identical eligible records, retaining both raw and 0–72-clipped outputs. The numerical source tables distinguish the primary nested procedure from fixed-structure, BSA-only and molecular-only comparisons. Published p-EASI uses serum alone, whereas the present procedure includes clinical BSA; local IL2RA is an unbridged proxy for sIL-2R. These local applications assess transport under unverified assay compatibility and do not establish model superiority or an ethnicity-specific difference.[11–13]

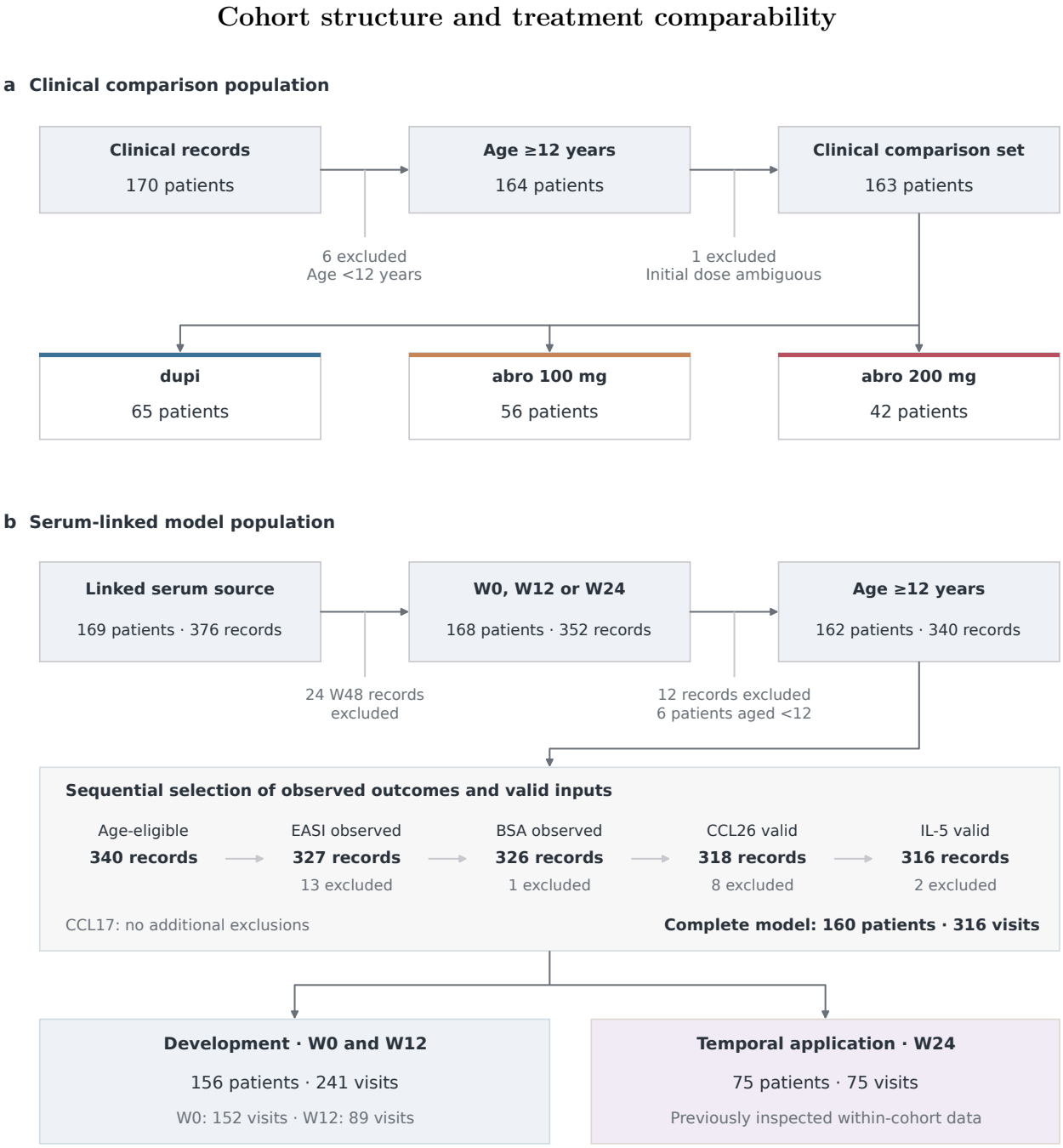


Figure S1: **Patient and record flow into the clinical and model analyses.** a, The curated clinical source contains 170 patients, not a screened or consented recruitment denominator. Six patients aged <12 years and one age-eligible patient with an ambiguous initial dose are excluded, leaving 163 patients classified by initial treatment: dupi, 65; abro 100 mg, 56; and abro 200 mg, 42. b, The linked serum source is a separate entry branch; model eligibility is not conditional on membership of the clinical comparison population. Removing 24 W48 records leaves 168 patients with 352 W0/W12/W24 records; age exclusions remove 12 records from six patients. Further ordered exclusions are mutually exclusive at the record level: 13 EASI, one BSA, zero CCL17, eight CCL26 and two IL-5 inputs are unavailable, leaving 316 complete model visits from 160 patients. A valid molecular input requires an available positive concentration for the log2 transformation. Loss of a record does not necessarily remove a patient. Development uses 241 W0/W12 visits from 156 patients, with patient-level validation splits; W24 includes 75 visits from 75 patients and is previously inspected within-cohort temporal application, not external validation. Development and W24 share 71 patients, with 85 development-only and four W24-only patients. Fifty-one patients have all three complete visits. The clinical and complete-input model populations share 159 patients, with four clinical-only and one model-only patient; the treatment-free model can retain an ambiguous initial dose. Scheduled empty visit slots are not counted as observed visits, and no filled-in clinical outcome is used as validation truth.

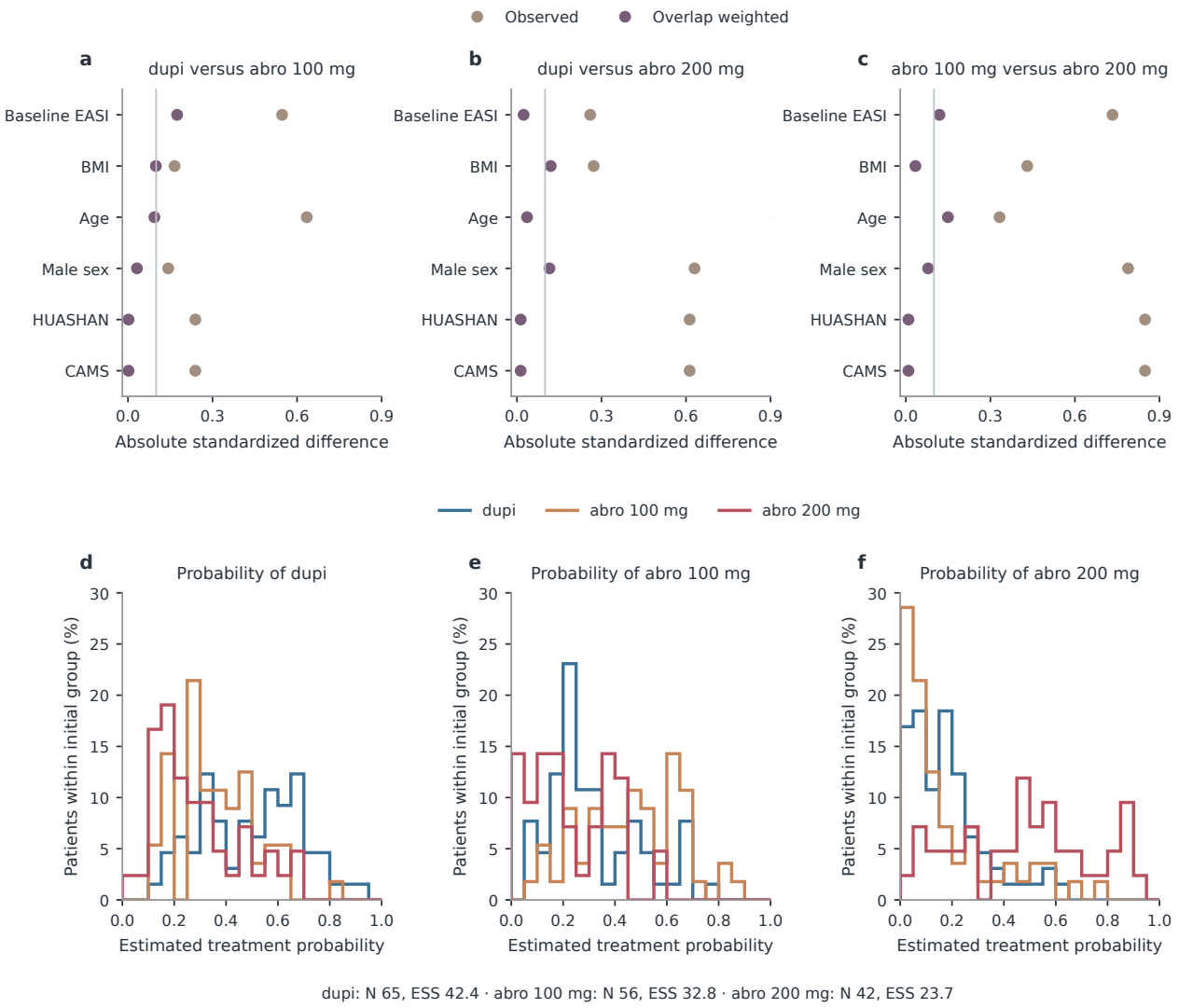


Figure S2: **Baseline balance, treatment overlap and effective sample size.** a–c, Absolute standardized mean differences for every covariate in the initial-treatment propensity model, before and after generalized overlap weighting. Differences use the unweighted pairwise pooled standard deviation in both displays. The light 0.10 reference line is a descriptive balance guide, not a statistical significance threshold. Residual imbalance remains. d–f, Baseline multinomial treatment probabilities, stratified by actual initial treatment; each distribution uses all patients in that initial group and sums to 100%. These show common support rather than longitudinal drug exposure. The footer reports original N and effective sample size, $(\text{sum w})^2/\text{sum}(\text{w}^2)$. All centre, sex, age, BMI and EASI variables are shown. Overlap weighting targets comparable baseline patients; it cannot remove unmeasured confounding or correct outcome-driven dose changes. Endpoint- and marker-specific availability/ESS diagnostics are retained in the source tables.

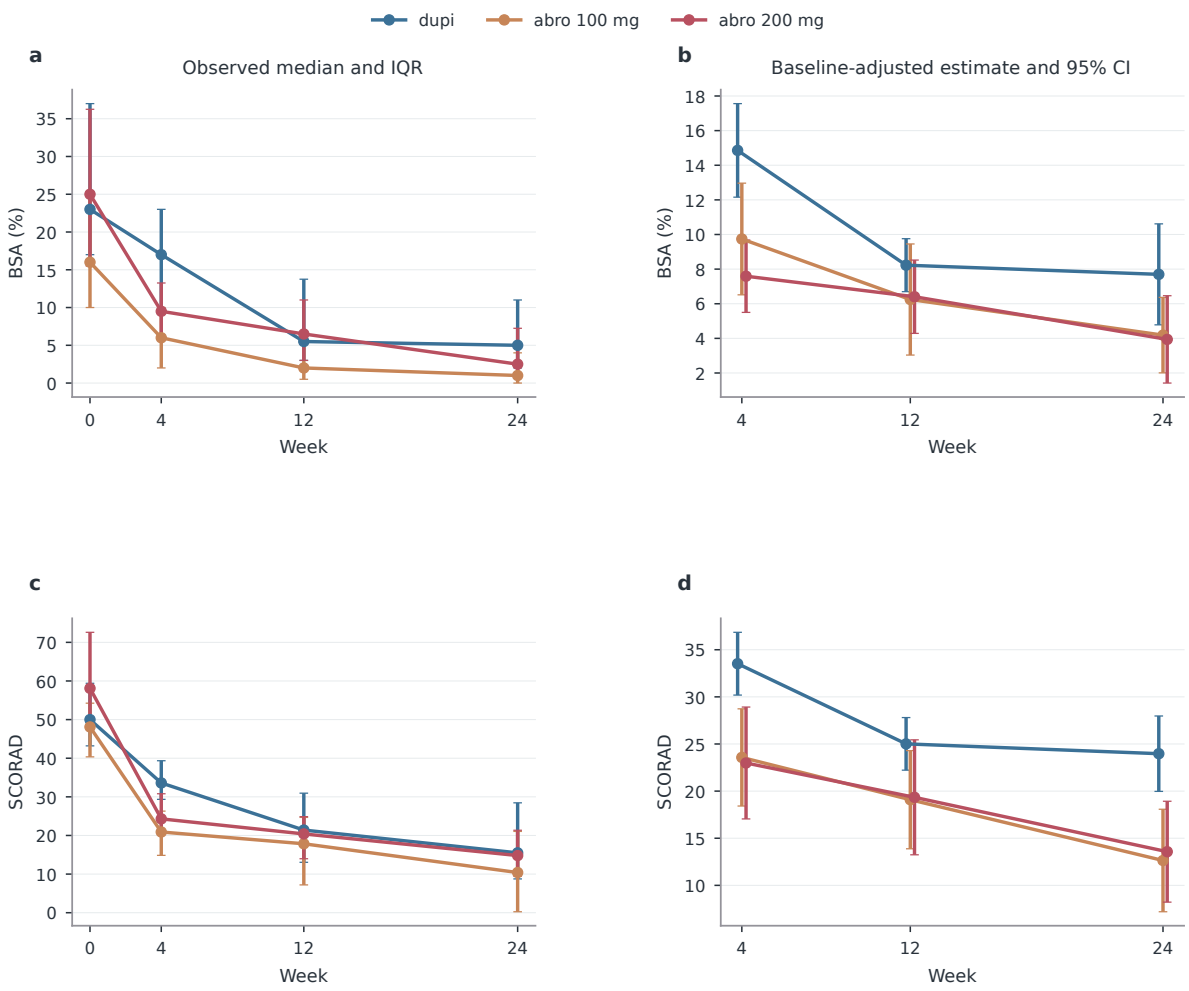


Figure S3: **Observed and adjusted BSA (%), SCORAD.** Left, observed medians with interquartile ranges by initial treatment; all available values are used at each specified visit. Right, generalized-overlap-weighted marginal estimates with patient-cluster robust 95% confidence intervals conditional on the estimated baseline weights, adjusted for baseline outcome, age, sex, BMI and centre. Patient sets can differ between visits. Exact outcome-specific counts and numerical comparisons are in the source tables; they are not replaced by a common denominator. Initial treatment groups describe observational management strategies, including subsequent regimen changes, rather than randomized fixed-dose effects.

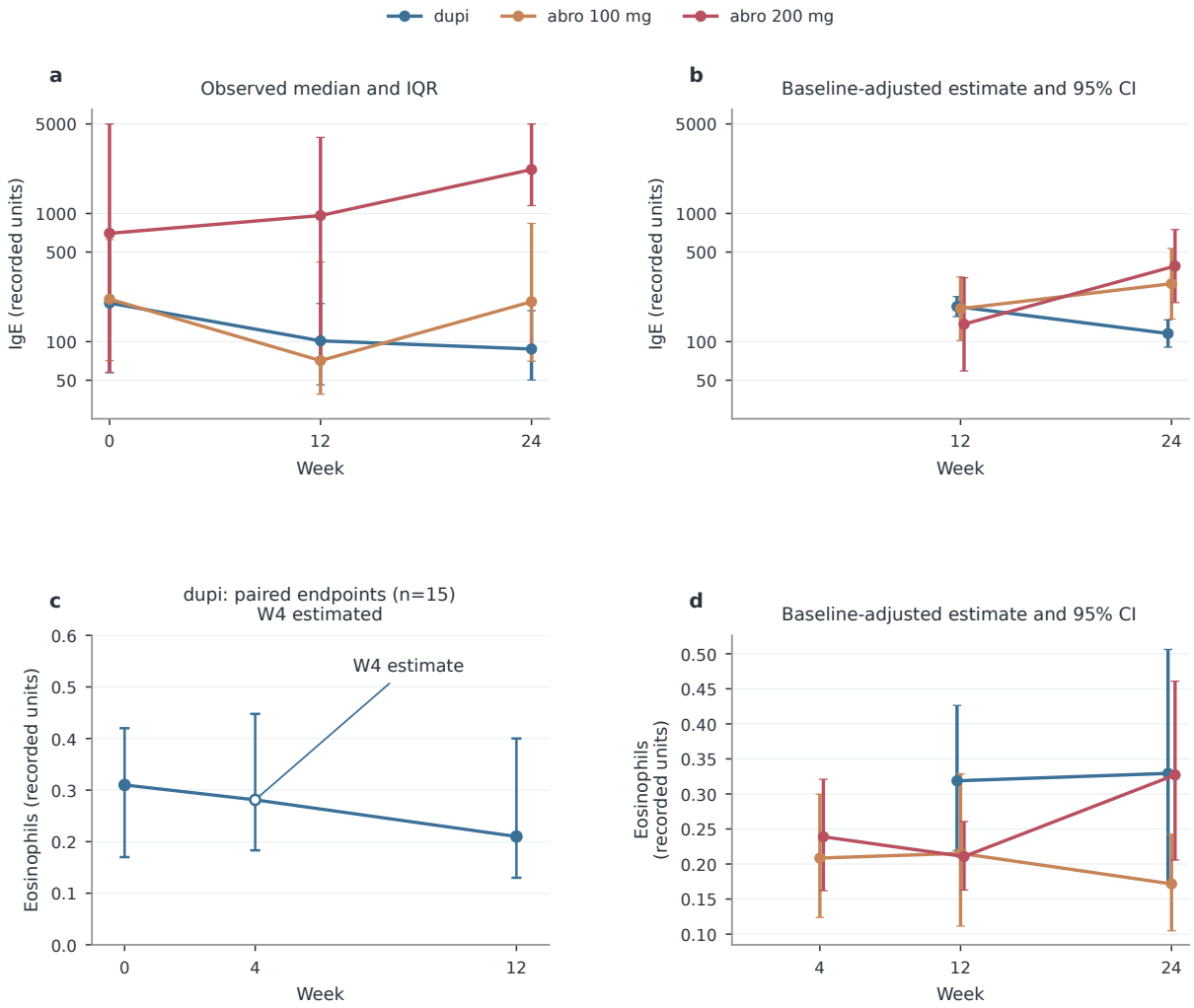


Figure S4: **IgE follow-up and the unobserved W4 eosinophil value.** a, Available-case IgE medians and interquartile ranges; patient sets vary by visit. The vertical axis is logarithmic with labels in recorded units. b,d, Generalized-overlap-weighted estimates from log(1+x) models adjusted for baseline outcome, age, sex, BMI and centre, with patient-cluster 95% confidence intervals conditional on estimated weights. Back-transformed estimates are not arithmetic means. There are no observed dupi W4 eosinophil measurements, so d contains no estimate for that treatment/visit cell. c, A separate fixed set of 15 dupi patients with W0 and W12 eosinophil measurements: filled points are observed endpoint medians; the hollow W4 point is an assumption-based estimate. Each patient's log(1+x) eosinophil value is linearly interpolated one-third of the way from W0 to W12, back-transformed, then summarized. Bars are patient-bootstrap 95% intervals for the median, conditional on this temporal assumption. The W4 estimate is 0.281 (0.183–0.448); it is not an observed result, validation outcome or treatment-comparison input. Original-scale interpolation, alternative time shapes and a seven-patient complete three-visit analysis are supplied separately. Initial treatment groups retain subsequent regimen changes.

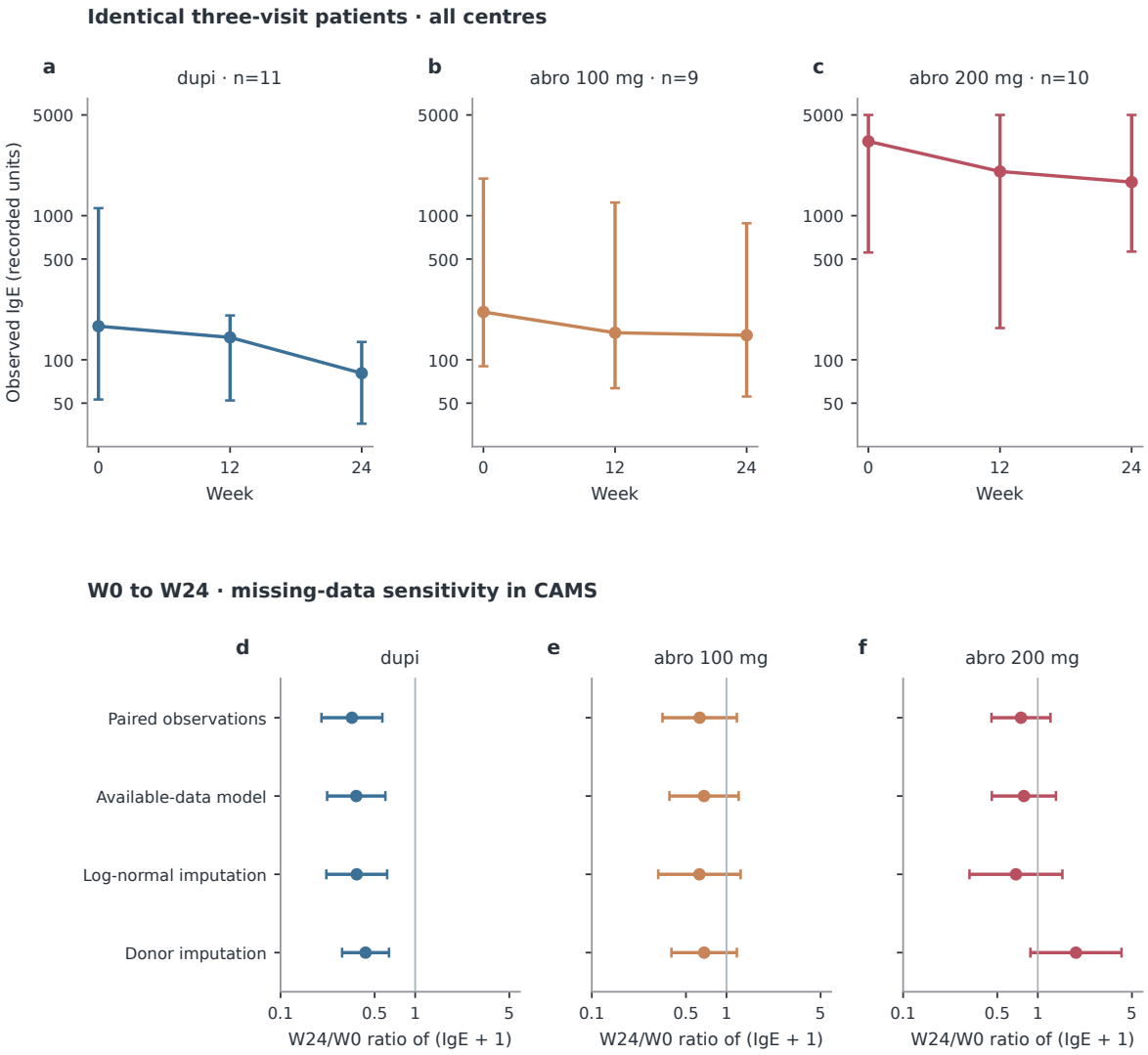


Figure S5: **IgE: fixed-patient observations and missing-data sensitivity.** a–c, Observed medians and patient-bootstrap 95% confidence intervals in the same complete three-visit patients at every visit: 11 dupi, 9 abro 100 mg and 10 abro 200 mg patients across both centres. The vertical scale is logarithmic, labelled in recorded units. d–f, W0–W24 geometric mean ratios of IgE+1 from four analyses within CAMS. The observed paired analysis uses all available W0/W24 pairs (14/10/12 by treatment); it does not require W12. The available-data random-intercept model uses 229 measured visits from 125 patients across the three treatments. Multiple-imputation analyses target all 130 CAMS patients (51/38/41), using patient-wide chained equations with either posterior log-normal draws or within-treatment predictive mean matching (donor imputation); 400 retained datasets are pooled with Rubin variance and finite-sample degrees of freedom. Both assume missingness can be explained by the included observed information. The panels target the same direction of change but rely on different populations or missingness assumptions. Ratios below one indicate lower W24 values; the line at one denotes no change. Bars are 95% confidence intervals. No observed value was replaced. Sparse donor support makes the abro 200 mg estimate sensitive to the imputation method; neither imputation confirms its direction of change. CAMS is used because Huashan has no follow-up IgE for dupi. Full-cohort, other-window and missing-not-at-random analyses are retained in Supplementary Table S8 and source data. No W4 IgE was observed, so no W4 point is imputed or plotted. The recorded value 5000 was not assumed to be a verified censoring threshold. These observational analyses do not identify causal drug effects.

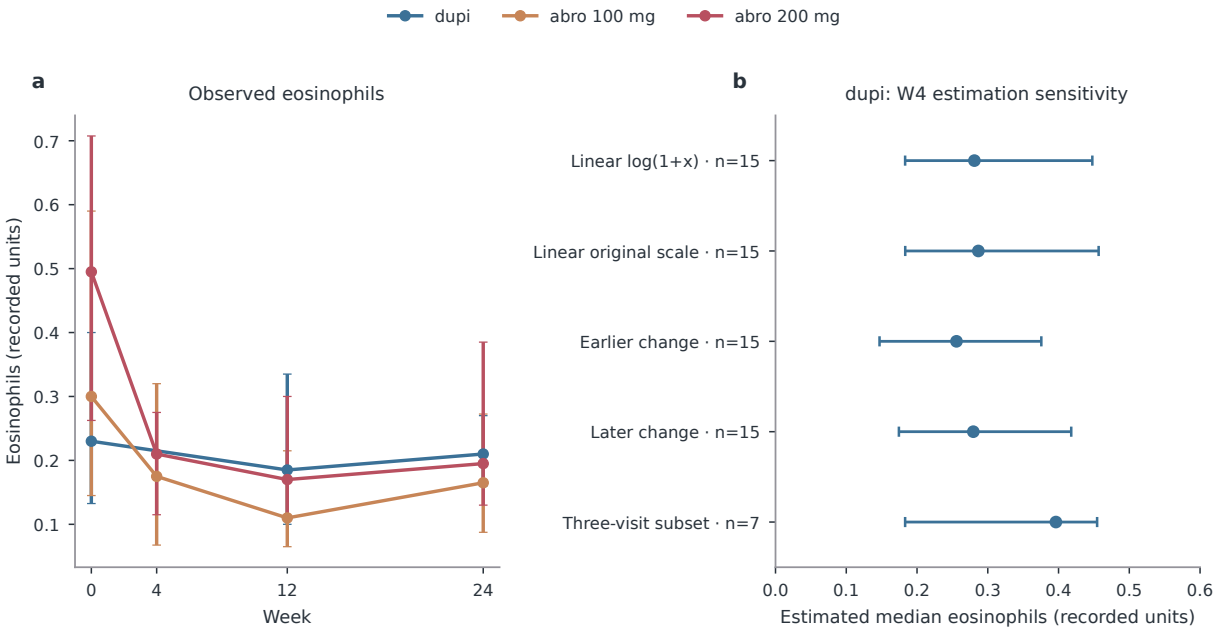


Figure S6: **Observed eosinophil data and sensitivity of the W4 estimate.** a, Available-case medians and interquartile ranges at observed visits. The dupi line connects W0 to W12 because no W4 measurement exists; crossing the W4 position does not create an observation. No imputed value enters this panel. b, Estimated dupi W4 medians and patient-bootstrap 95% intervals under different temporal assumptions. The first four estimates use the same 15 W0/W12 pairs. Earlier/later change replaces the linear time fraction with $\log(1+t)/\log(13)$ or $(t/12)^2$ at t=4. The last row restricts to seven patients also observed at W24, illustrating selection sensitivity, not a better estimate. Intervals condition on the stated assumptions and do not cover all uncertainty about the unmeasured W4 distribution. These are post hoc descriptive estimates, not multiple imputation under an identified treatment-specific W4 model. All original observations remain unchanged.

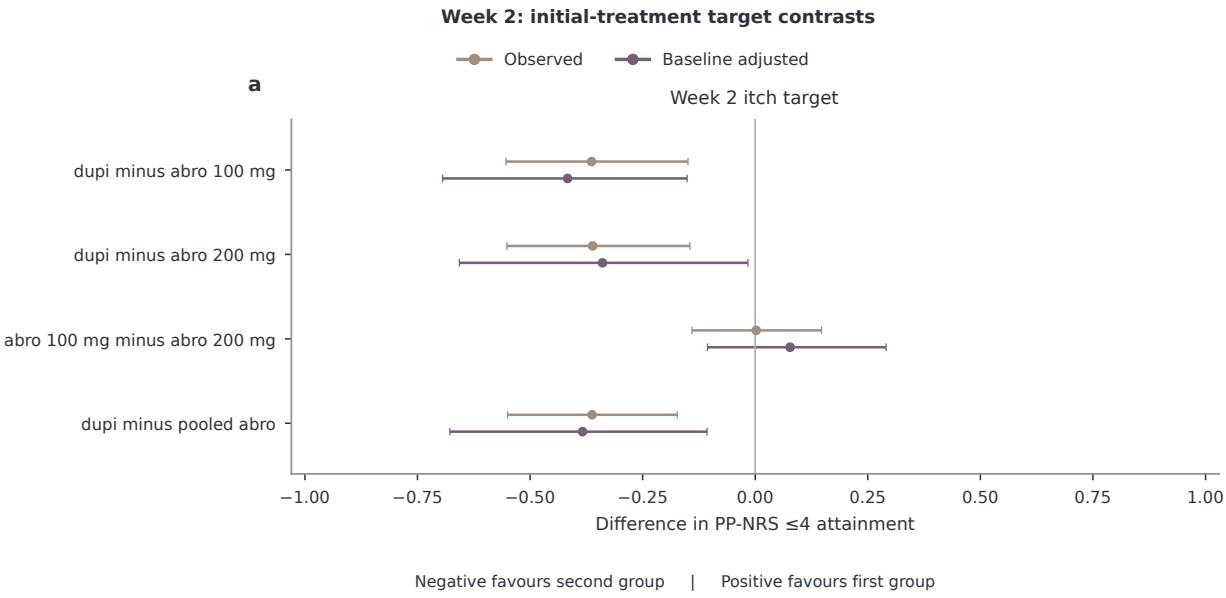


Figure S7: **Week 2: all initial-treatment target contrasts.** Observed and baseline-adjusted risk differences with nominal 95% confidence intervals. Positive values favour the first named treatment for target attainment. Adjustment uses generalized overlap weights from baseline EASI, age, sex, BMI and centre; it does not estimate a randomized current-dose effect. Treatment labels refer to the initial prescribed regimen. pooled abro is supplementary. Missing estimates are left absent; all exact denominators, Fisher/weighted tests, Holm-adjusted P values and exploratory FDR values are in the source tables.

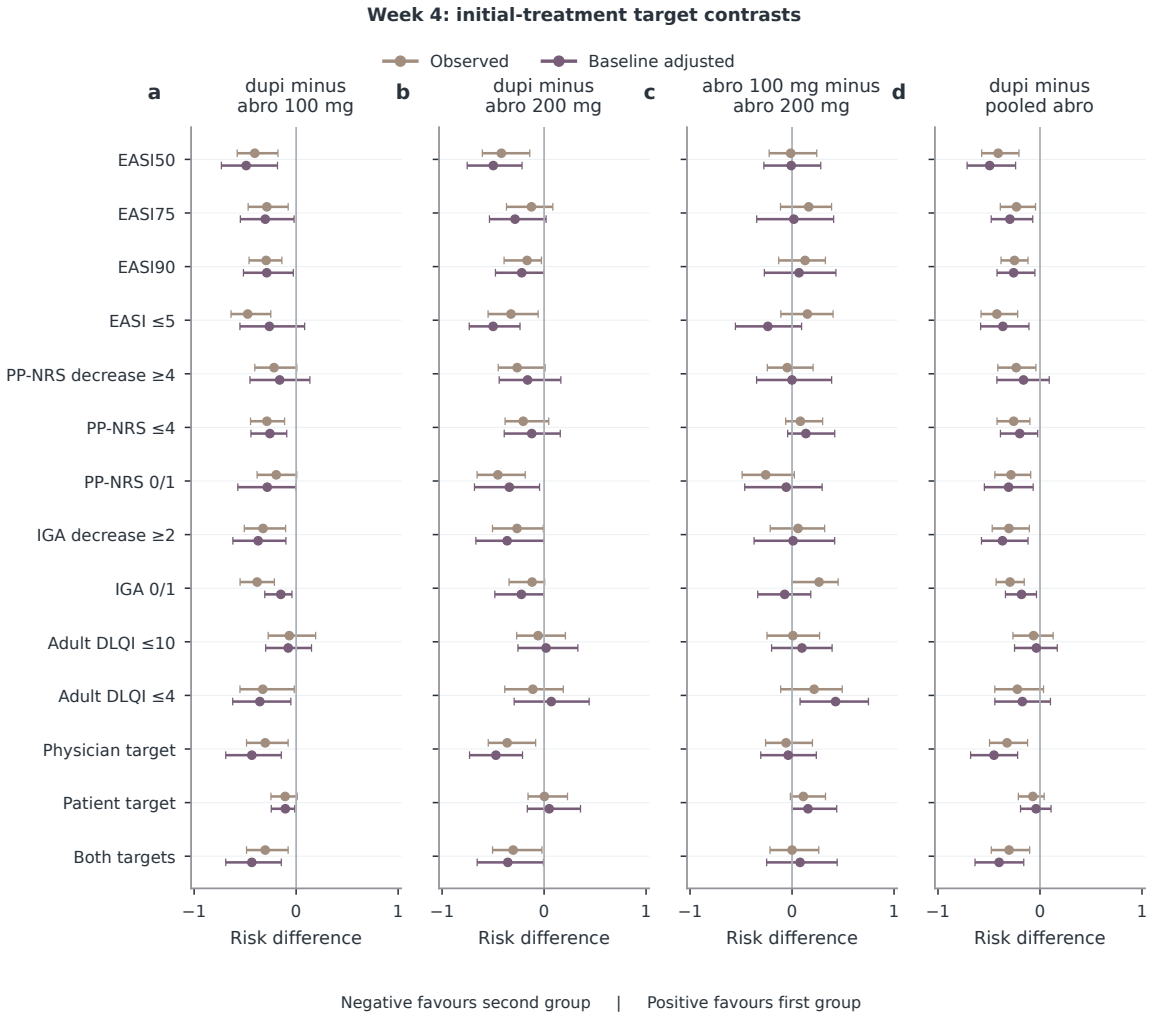


Figure S8: **Week 4: all initial-treatment target contrasts.** Observed and baseline-adjusted risk differences with nominal 95% confidence intervals. Positive values favour the first named treatment for target attainment. Adjustment uses generalized overlap weights from baseline EASI, age, sex, BMI and centre; it does not estimate a randomized current-dose effect. Treatment labels refer to the initial prescribed regimen. pooled abro is supplementary. Missing estimates are left absent; all exact denominators, Fisher/weighted tests, Holm-adjusted P values and exploratory FDR values are in the source tables.

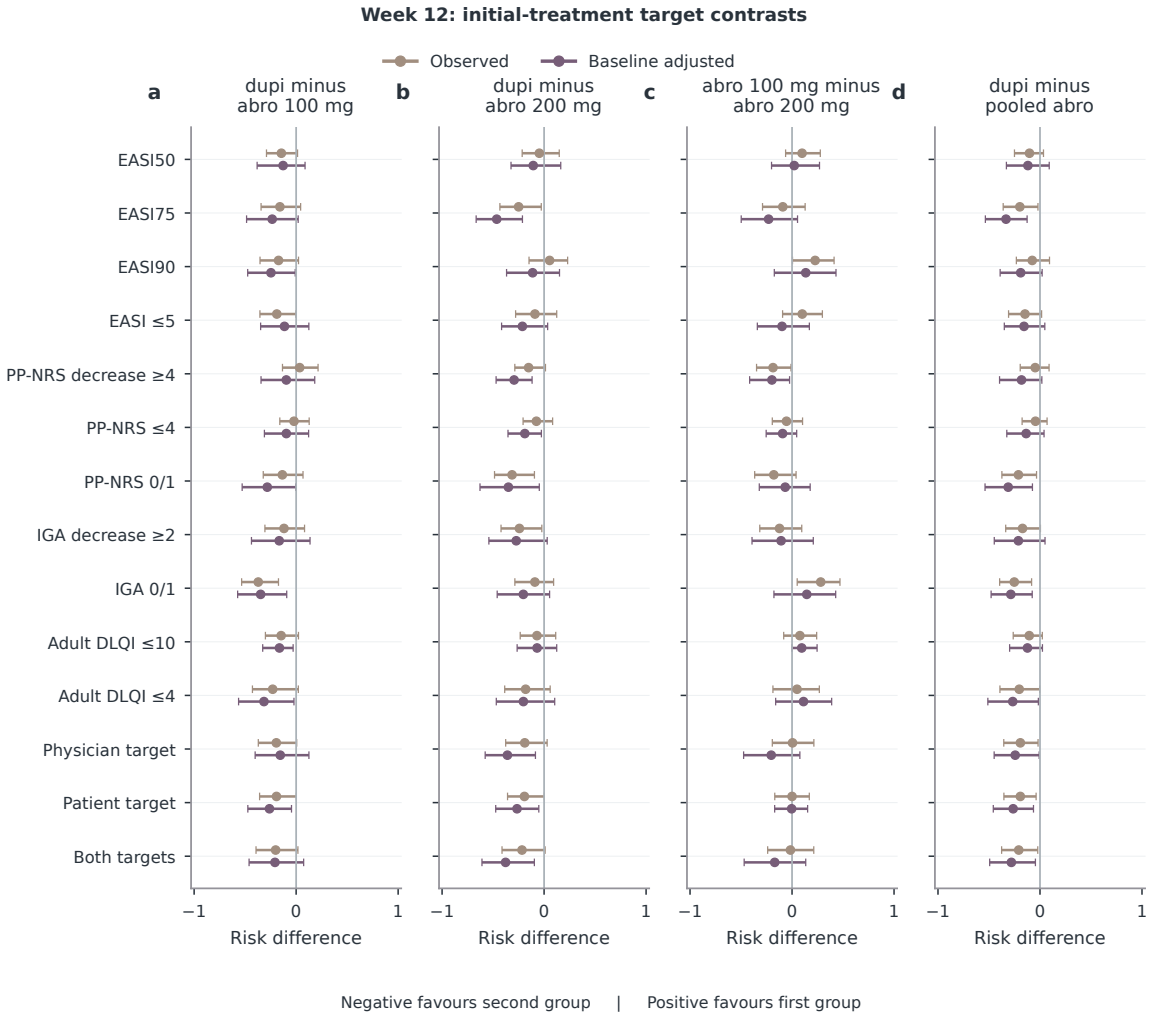


Figure S9: **Week 12: all initial-treatment target contrasts.** Observed and baseline-adjusted risk differences with nominal 95% confidence intervals. Positive values favour the first named treatment for target attainment. Adjustment uses generalized overlap weights from baseline EASI, age, sex, BMI and centre; it does not estimate a randomized current-dose effect. Treatment labels refer to the initial prescribed regimen. pooled abro is supplementary. Missing estimates are left absent; all exact denominators, Fisher/weighted tests, Holm-adjusted P values and exploratory FDR values are in the source tables.

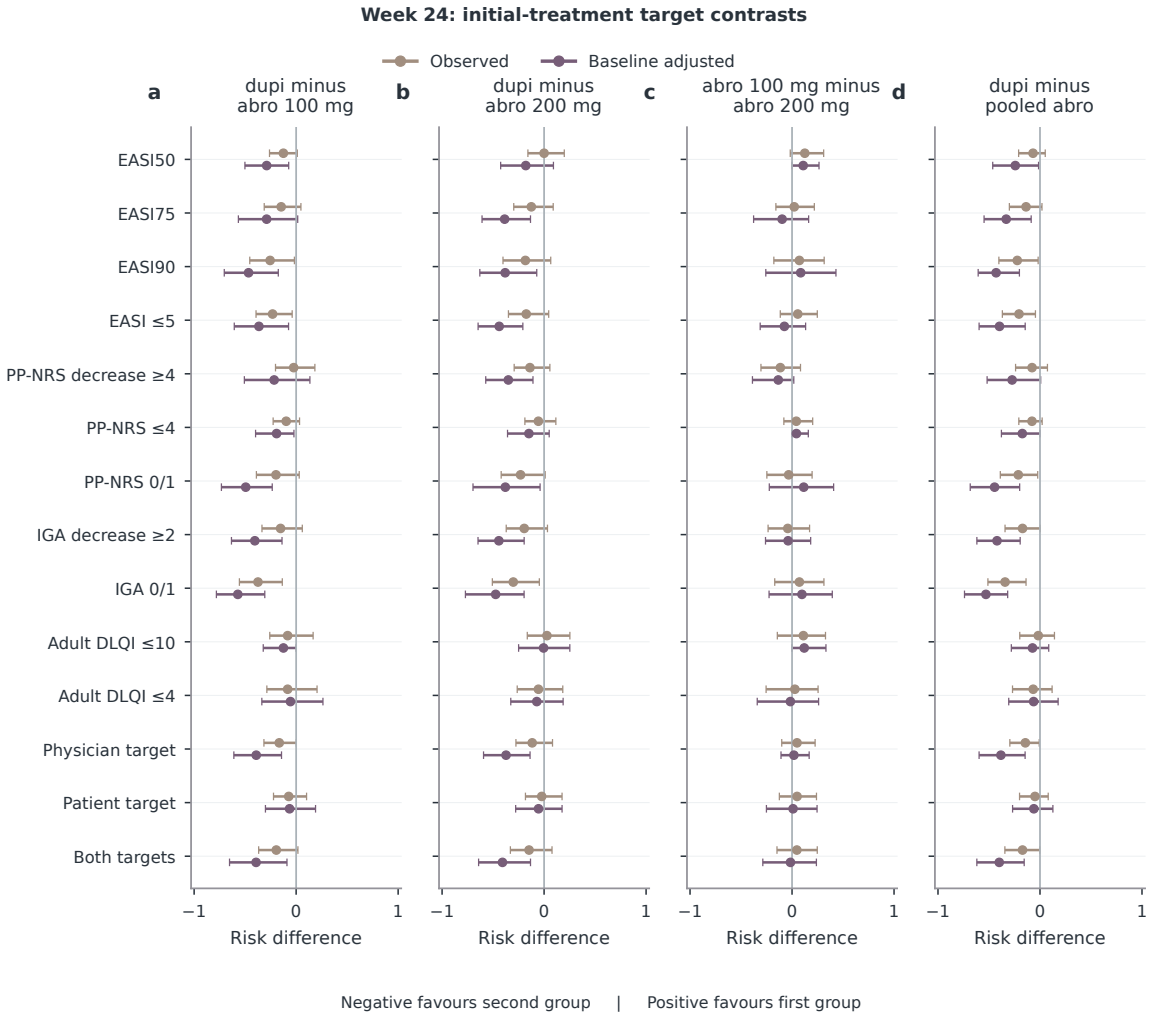


Figure S10: **Week 24: all initial-treatment target contrasts.** Observed and baseline-adjusted risk differences with nominal 95% confidence intervals. Positive values favour the first named treatment for target attainment. Adjustment uses generalized overlap weights from baseline EASI, age, sex, BMI and centre; it does not estimate a randomized current-dose effect. Treatment labels refer to the initial prescribed regimen. pooled abro is supplementary. Missing estimates are left absent; all exact denominators, Fisher/weighted tests, Holm-adjusted P values and exploratory FDR values are in the source tables.

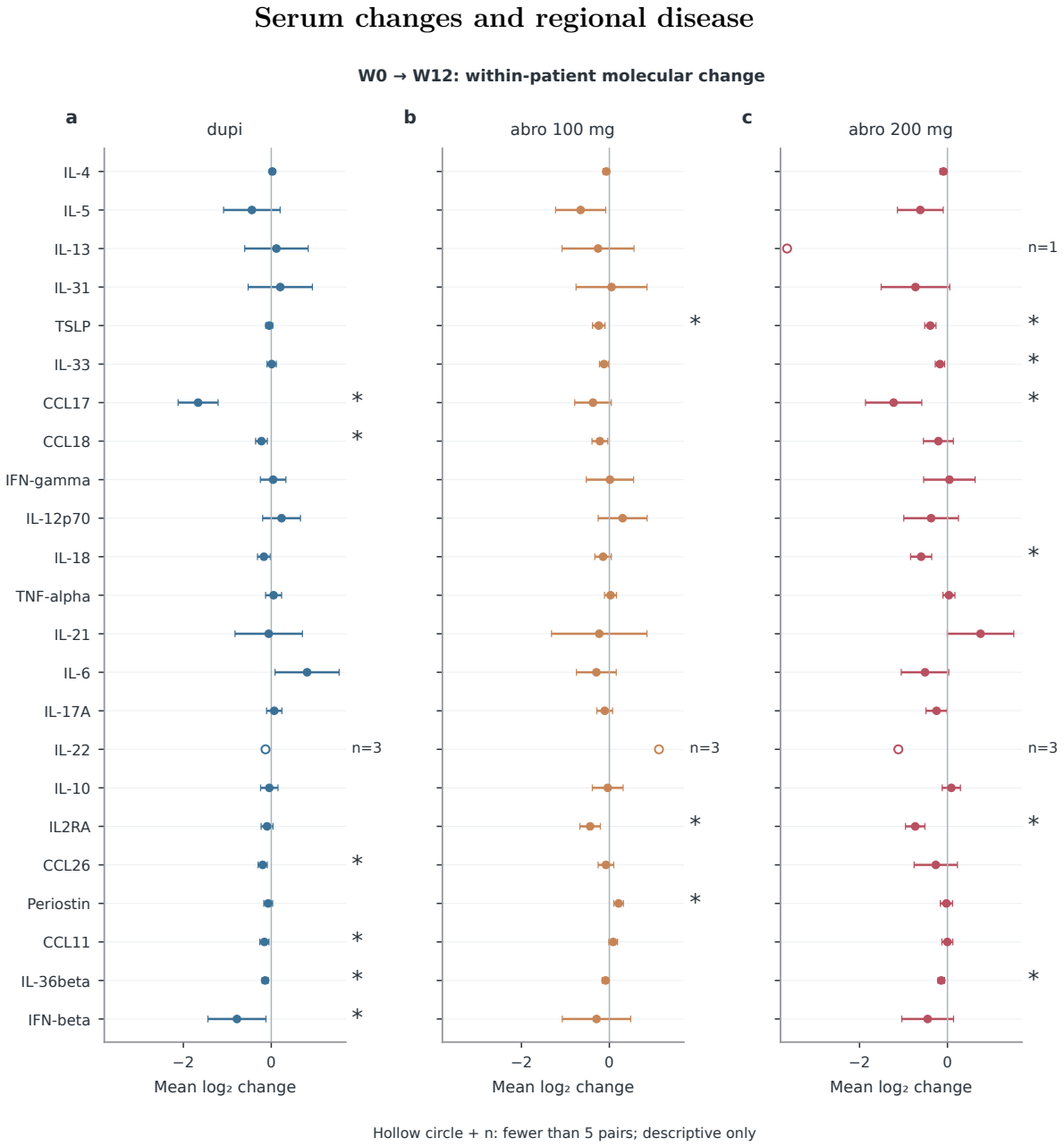


Figure S11: **Molecular changes from W0 to W12.** Mean paired $\log_2$ concentration changes with 95% t confidence intervals. Negative values indicate lower later concentration. Each marker/treatment uses all available pairs for this interval; exact counts are in the source table. All 23 measured markers are retained. Hollow circles with n labels denote fewer than five pairs; these are descriptive and have no inferential interval or star. Unavailable estimates are blank. Stars denote BH $q<0.05$ across the 193 estimable tests among 207 planned marker-by-treatment-by-window comparisons, not a separate family for the displayed interval.

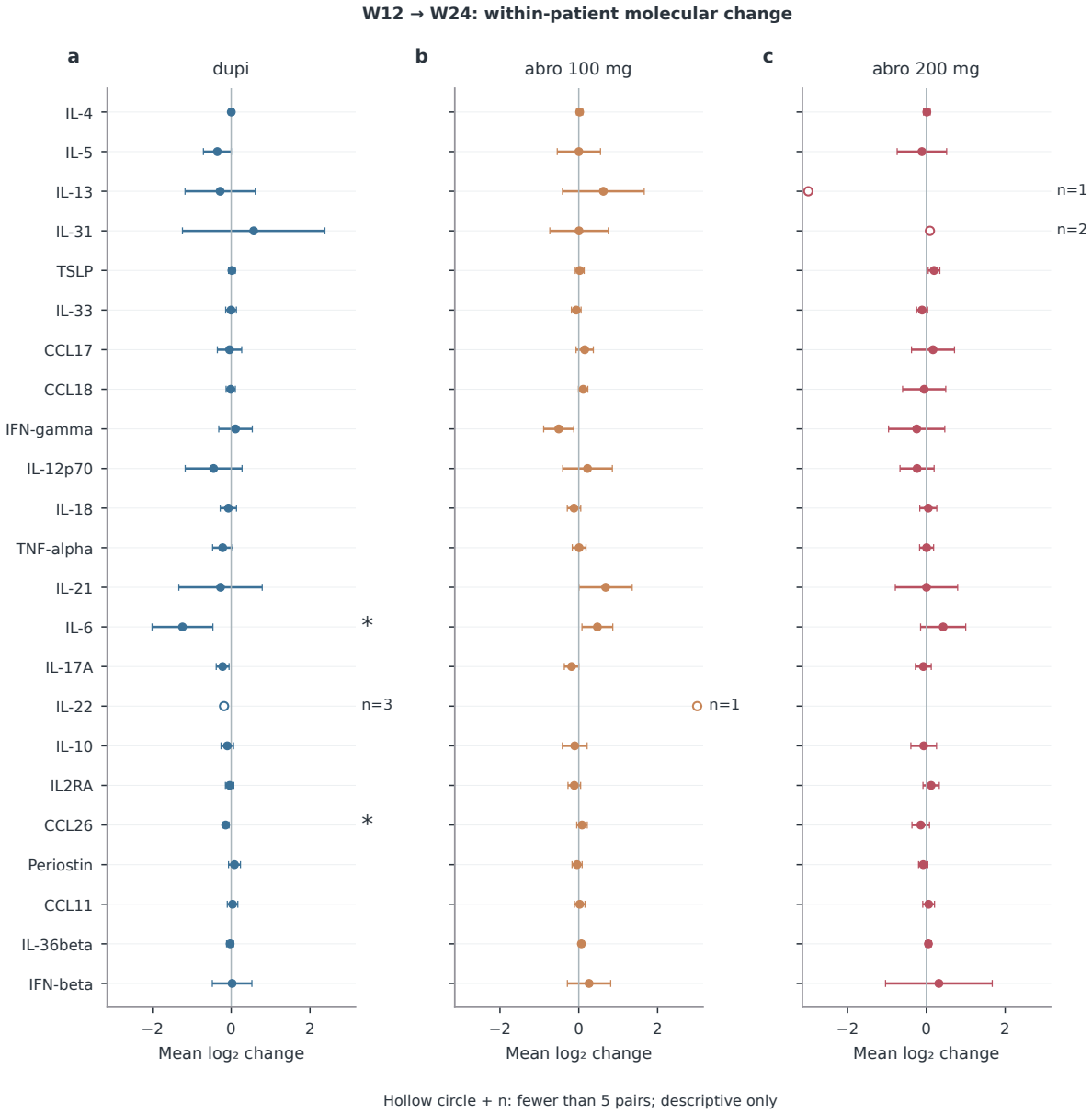


Figure S12: **Molecular changes from W12 to W24.** Mean paired $\log_2$ concentration changes with 95% t confidence intervals. Negative values indicate lower later concentration. Each marker/treatment uses all available pairs for this interval; exact counts are in the source table. All 23 measured markers are retained. Hollow circles with n labels denote fewer than five pairs; these are descriptive and have no inferential interval or star. Unavailable estimates are blank. Stars denote BH q<0.05 across the 193 estimable tests among 207 planned marker-by-treatment-by-window comparisons, not a separate family for the displayed interval.

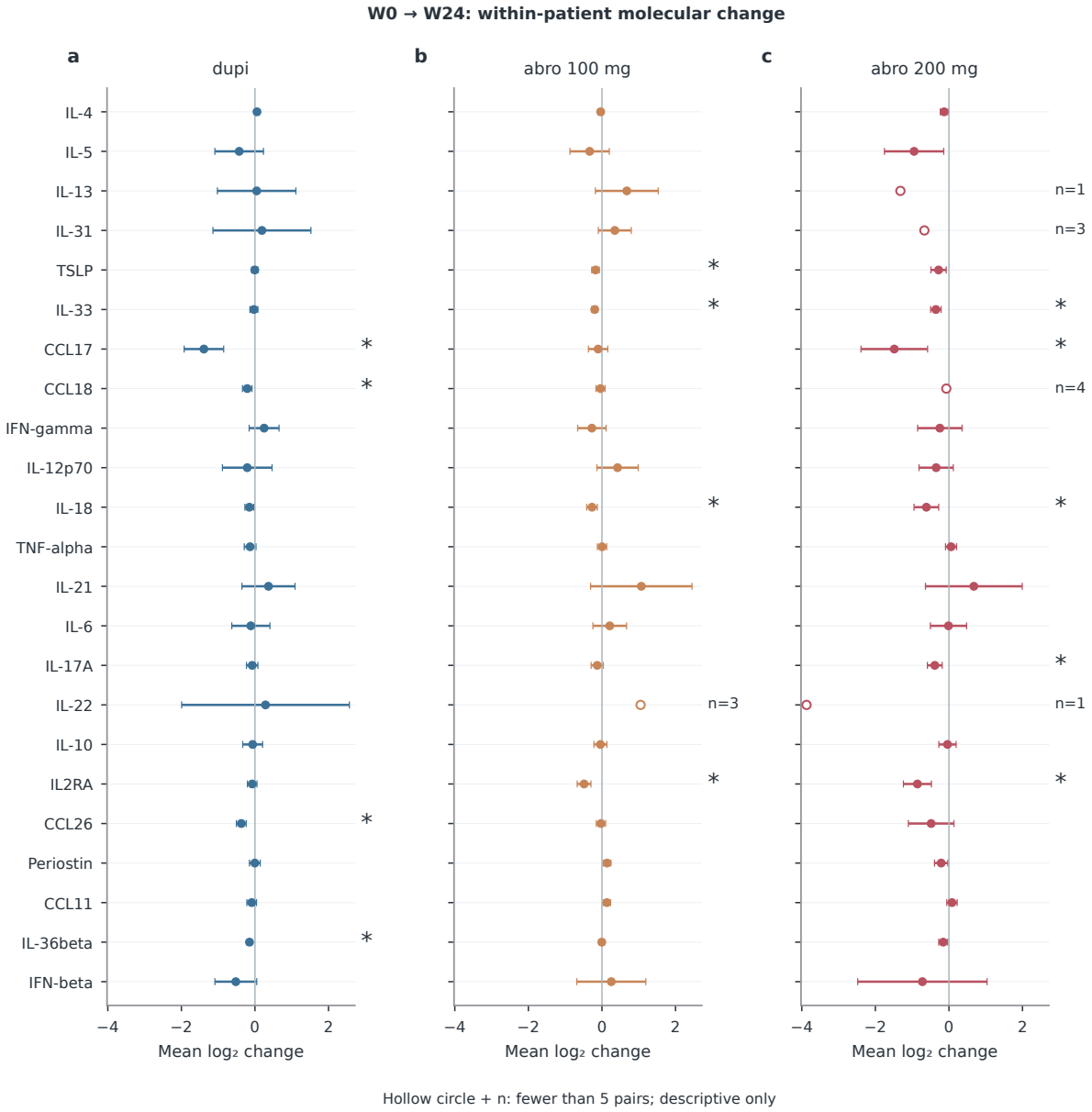


Figure S13: **Molecular changes from W0 to W24.** Mean paired $\log_2$ concentration changes with 95% t confidence intervals. Negative values indicate lower later concentration. Each marker/treatment uses all available pairs for this interval; exact counts are in the source table. All 23 measured markers are retained. Hollow circles with n labels denote fewer than five pairs; these are descriptive and have no inferential interval or star. Unavailable estimates are blank. Stars denote BH $q<0.05$ across the 193 estimable tests among 207 planned marker-by-treatment-by-window comparisons, not a separate family for the displayed interval.

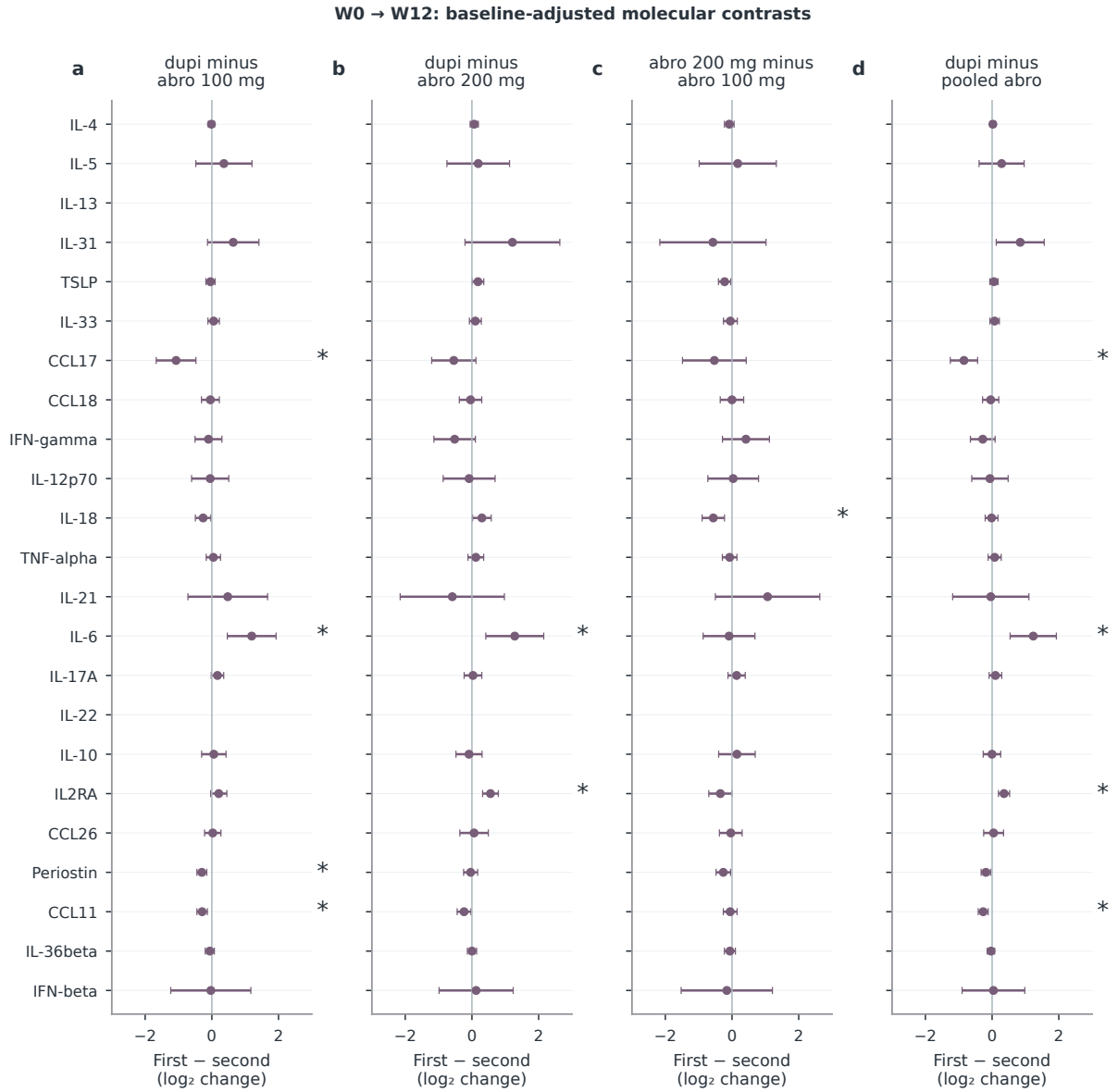


Figure S14: **Adjusted molecular comparisons from W0 to W12.** Differences in within-patient $\log_2$ changes with HC3 95% confidence intervals. Negative values mean a smaller, more negative change in the first named treatment. Models adjust starting $\log_2$ concentration, baseline total EASI, age, sex, BMI and centre. dupi denotes dupilumab; abro denotes abrocitinib, with 100/200 mg identifying initial dose. pooled abro retains the observed dose mixture. Stars denote the original BH q<0.05 family; no multiplicity family is recalculated after display selection. Blank positions have no estimable contrast. Complete counts, diagnostics, centre/batch sensitivities and tests remain in the source table. These are adjusted observational associations, not randomized current-dose effects.

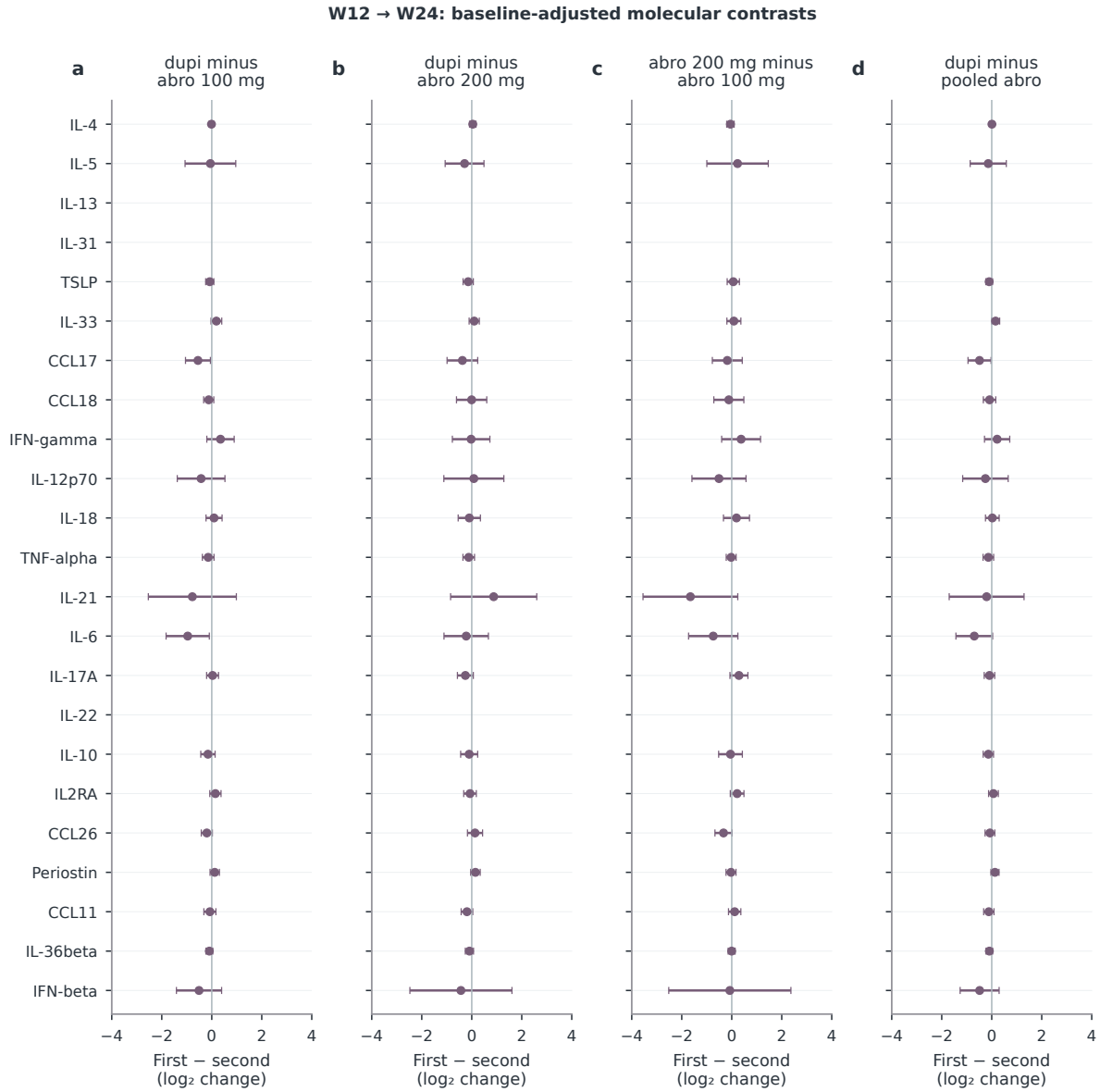


Figure S15: **Adjusted molecular comparisons from W12 to W24.** Differences in within-patient $\log_2$ changes with HC3 95% confidence intervals. Negative values mean a smaller, more negative change in the first named treatment. Models adjust starting $\log_2$ concentration, baseline total EASI, age, sex, BMI and centre. dupi denotes dupilumab; abro denotes abrocitinib, with 100/200 mg identifying initial dose. pooled abro retains the observed dose mixture. Stars denote the original BH $q<0.05$ family; no multiplicity family is recalculated after display selection. Blank positions have no estimable contrast. Complete counts, diagnostics, centre/batch sensitivities and tests remain in the source table. These are adjusted observational associations, not randomized current-dose effects.

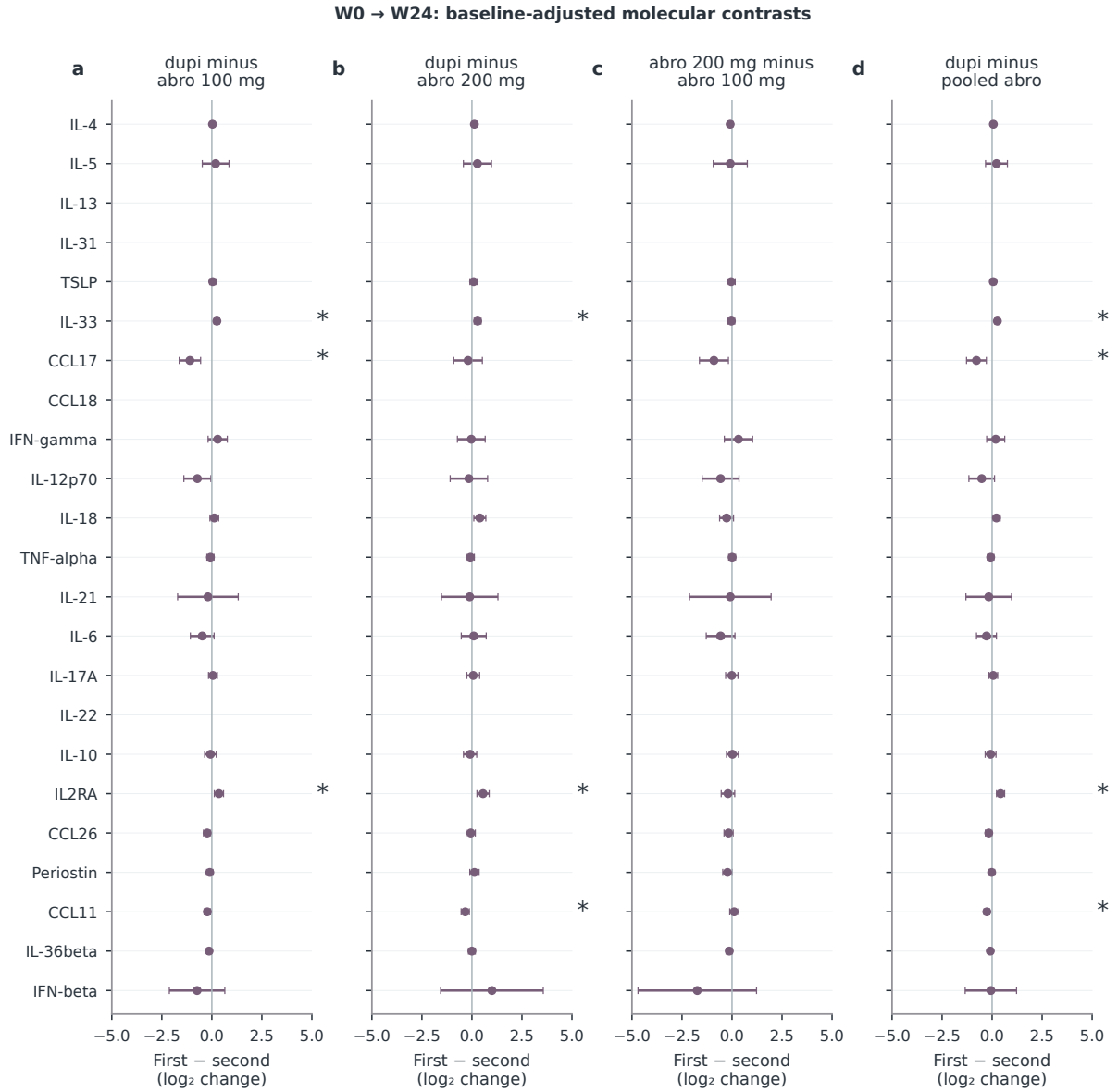


Figure S16: **Adjusted molecular comparisons from W0 to W24.** Differences in within-patient $\log_2$ changes with HC3 95% confidence intervals. Negative values mean a smaller, more negative change in the first named treatment. Models adjust starting $\log_2$ concentration, baseline total EASI, age, sex, BMI and centre. dupi denotes dupilumab; abro denotes abrocitinib, with 100/200 mg identifying initial dose. pooled abro retains the observed dose mixture. Stars denote the original BH q<0.05 family; no multiplicity family is recalculated after display selection. Blank positions have no estimable contrast. Complete counts, diagnostics, centre/batch sensitivities and tests remain in the source table. These are adjusted observational associations, not randomized current-dose effects.

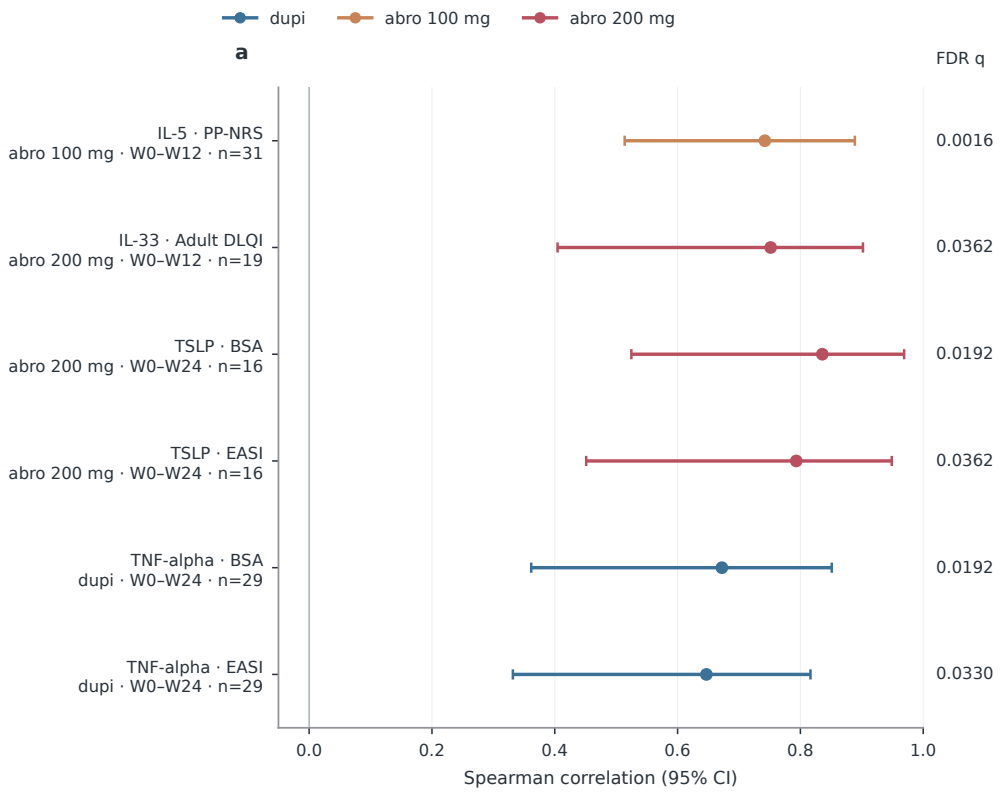


Figure S17: **Molecular and clinical changes associated after multiplicity correction.** Every association meeting BH q<0.05 in the full declared change-association family is displayed; selection spans 23 markers, five clinical outcomes, three initial treatments and three intervals. No W12–W24 association met this threshold. Points are Spearman correlations between within-patient clinical change and $\log_2$ molecular change, both later minus earlier, with patient-bootstrap 95% confidence intervals. Positive correlations indicate changes in the same direction. Labels give paired-patient counts and the interval; colours indicate initial treatment only. Adult DLQI requires age ≥18. All tested coefficients, non-significant results and unestimable pairs remain in the complete source table. This significance-selected display is exploratory and is not independent validation, numerical agreement or a causal molecular effect.

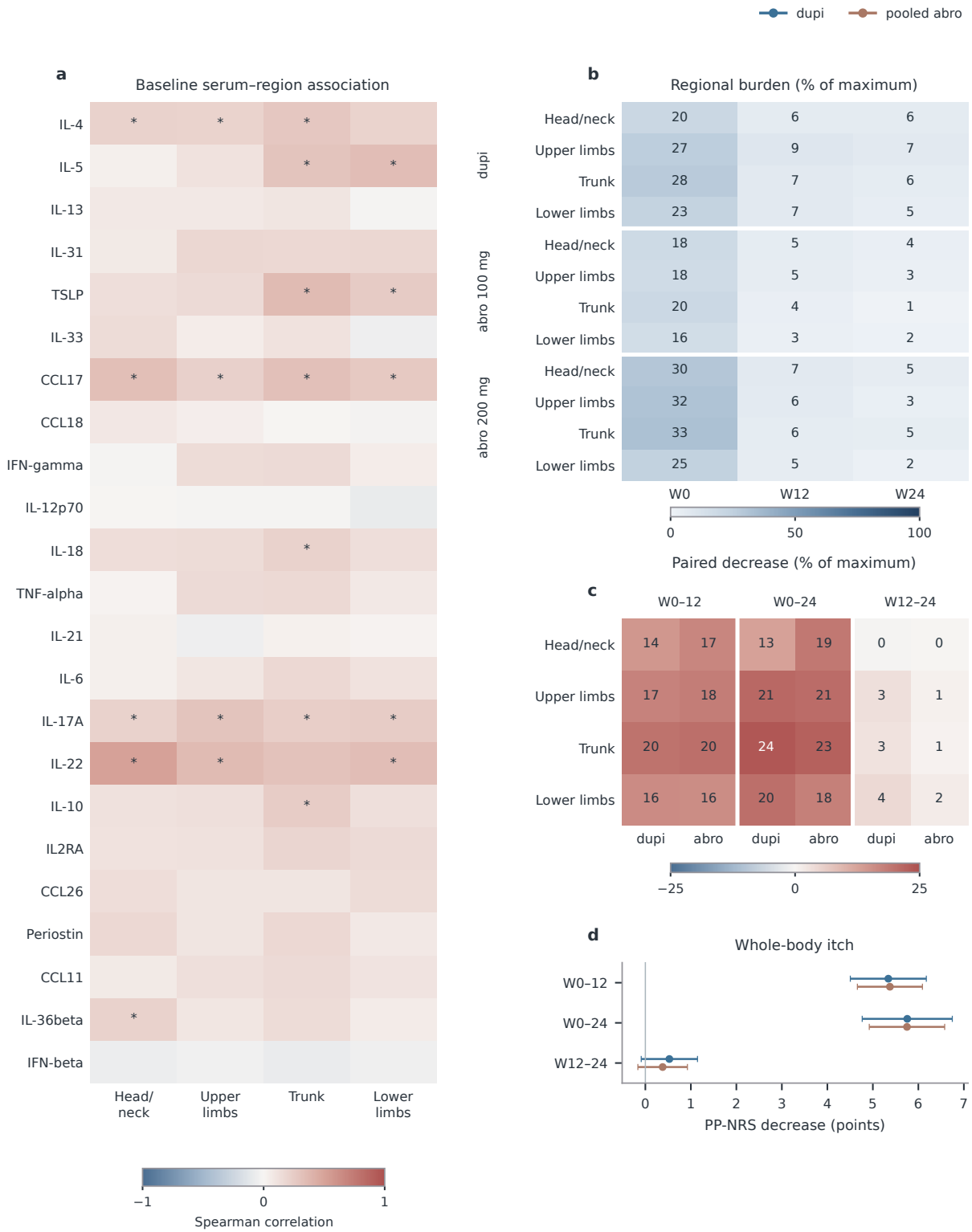


Figure S18: **Regional clinical burden and molecular associations.** a, Baseline Spearman associations between all 23 markers and the four regional EASI contributions; stars mark BH q<0.05, with complete coefficients, patient counts and bootstrap intervals retained in the source. b, Available-case mean regional EASI divided by its theoretical maximum and multiplied by 100; cells show percentages, not patient counts. c, Mean paired regional decrease as a percentage of the same regional maximum for three separately labelled intervals; positive values indicate improvement. The abro columns represent the observed pooled-dose mixture. These heatmaps summarize magnitude; exact 95% t intervals and counts are in the source tables. d, Whole-body PP-NRS mean decreases with 95% t intervals, not region-specific itch. Regional maxima are 7.2, 14.4, 21.6 and 28.8 for head/neck, upper limbs, trunk and lower limbs. All regional analyses require complete, bounded regional records consistent with total EASI. Intervals can share patients and are not successive cumulative effects. Colour bars encode distinct quantities and are not interchangeable. No baseline-adjusted regional treatment contrast passed the FDR threshold; full separate-dose comparisons remain available.

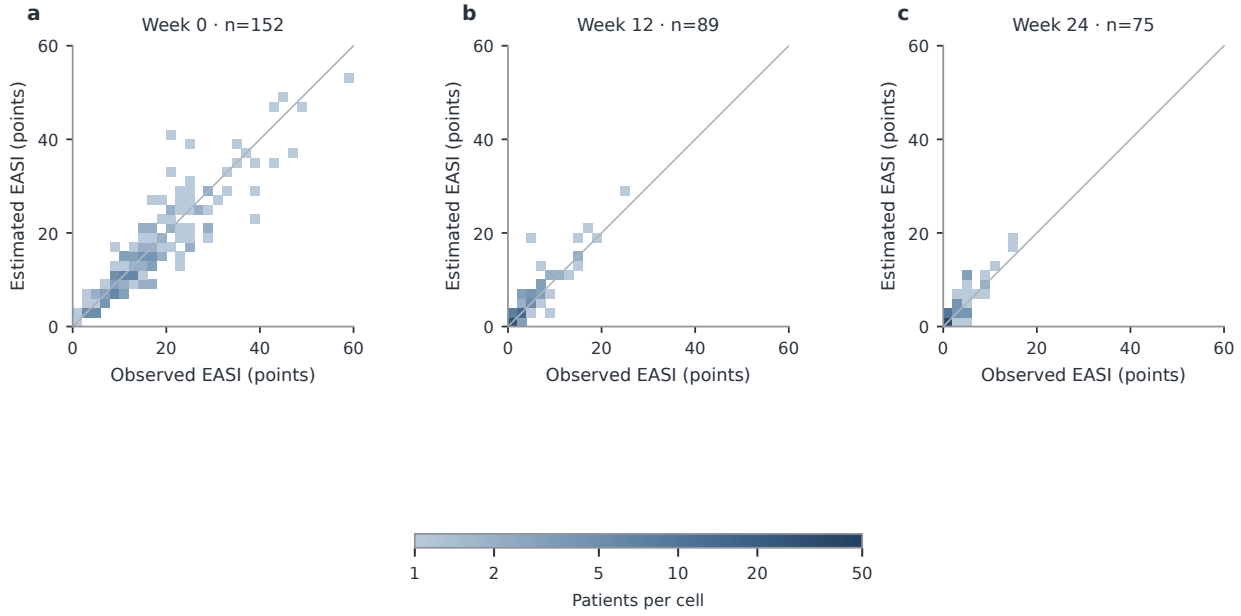


Figure S19: **Observed and estimated EASI at each visit.** a–c, Estimated versus observed EASI in the first patient-held-out repetition at W0/W12 and same-cohort temporal application at W24. Each fixed 2×2-EASI-point cell counts all exact score pairs within its boundaries. Colour gives patient count on one shared logarithmic scale; white cells contain no observations. All 152/89/75 eligible patients are retained. Identical axes preserve the severity-range comparison; grey diagonals denote exact agreement. W0/W12 use nested patient-level structure and penalty selection; earlier marker selection is not repeated. W24 had previously been inspected. These results assess the development procedure rather than independent validation of the final coefficient vector. Overall performance is in Table 2; the main figure displays the complete absolute-error distribution.

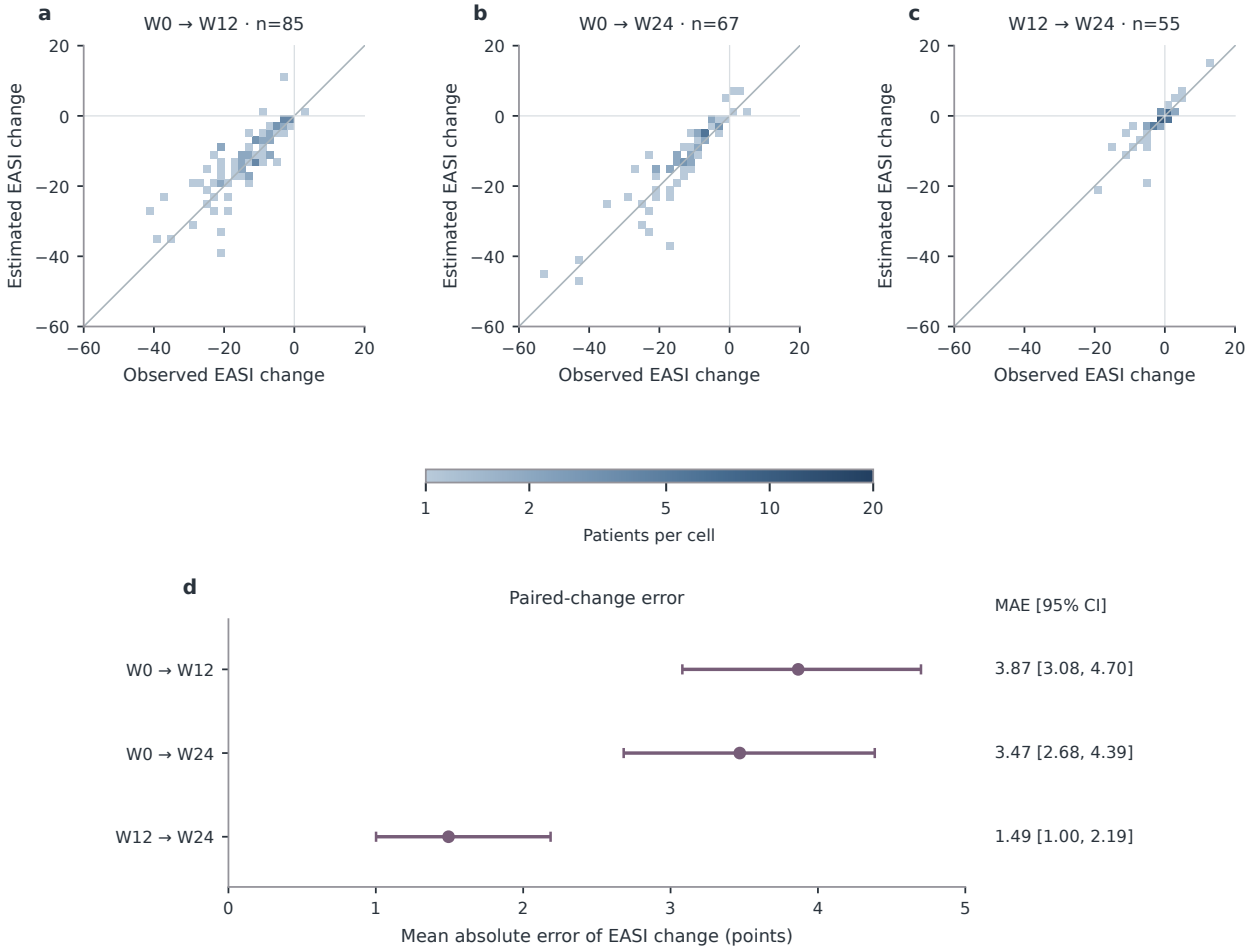


Figure S20: **Within-patient EASI change and absolute error.** a–c, Estimated versus observed EASI changes in the first patient-held-out repetition. Change is end minus start: negative values indicate improvement. Fixed 2×2-EASI-point cells count exact paired changes on one logarithmic count scale. Identical axes retain all 85, 67 and 55 pairs; the intervals can share patients and are not consecutive categories on a time axis. White cells contain no observations; diagonals mark exact agreement. d, Change MAE calculated within each of two repetitions before averaging, with 2,000 patient-bootstrap 95% confidence intervals conditional on saved predictions. Both scores are paired within each repetition before subtraction. W24 is previously inspected same-cohort temporal application. EASI75 classification is shown in the main model figure; complete EASI50/75/90 classifications are retained separately.

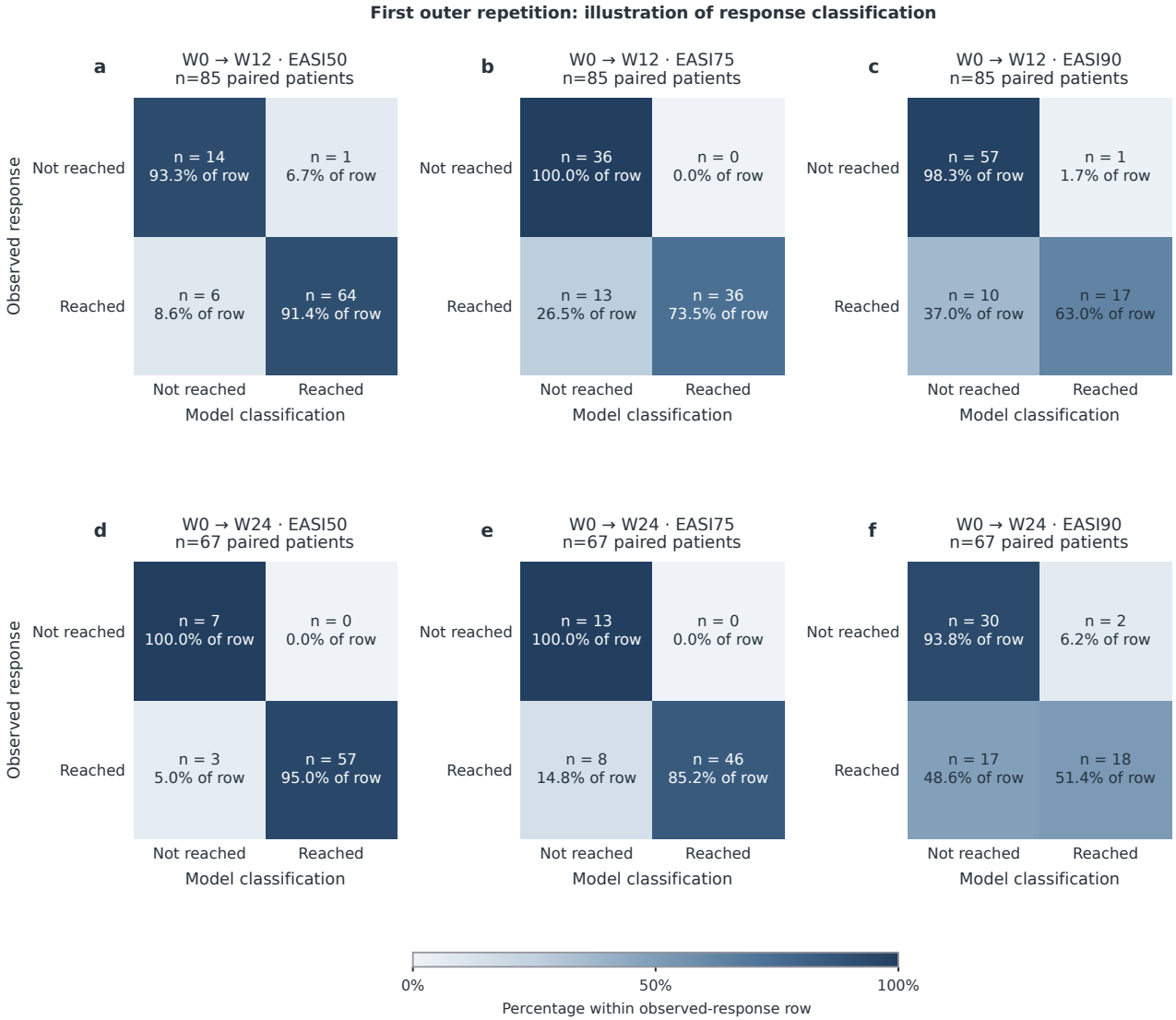


Figure S21: **Classification of EASI50, EASI75 and EASI90 response.** Each panel shows actual patient counts from the first outer repetition, separately for W0→W12 and W0→W24. Rows are observed response and columns model-estimated response. Diagonal cells are correct; upper-right cells falsely indicate response and lower-left cells miss an observed responder. Percentages are within each observed-response row, so the lower-right percentage is sensitivity and the upper-left percentage specificity. Observed response uses observed baseline/follow-up scores, whereas estimated response uses predicted baseline/follow-up scores. Counts are integers from one repetition, not the primary two-repetition average; the full metrics table retains the latter. The paired denominators are 85 and 67, with no excluded pair for invalid baseline in this displayed repetition. The two intervals can share patients. This assesses response classification, not alternative-treatment outcomes or continuous-score equivalence.

Aggregate source tables

The machine-readable source-table index lists the complete aggregate collection, including unplotted and negative results. Figure-specific dependencies are in `figure_catalog.json`. The versioned aggregate directory includes clinical, molecular, model, regional, matched-input and missing-data sensitivity tables; restricted patient-level files are not part of the public collection.

References

[1] Marjolein de Bruin-Weller et al. Treat-to-Target in Atopic Dermatitis: An International Consensus on a Set of Core Decision Points for Systemic Therapies. *Acta Dermato-Venereologica*, 101:adv00402, 2021. doi: 10.2340/00015555-3751.

[2] Zuotao Zhao and Xinghua Gao. Expert recommendations on treat-to-target in the systemic treatment of moderate to severe atopic dermatitis. *Chinese Journal of Dermatovenereology*, 36(8):855–864, 2022. doi: 10.13735/j.cjdv.1001-7089.202205115. In Chinese; thresholds cross-checked against the authors' 2026 clinical-trial analysis.

[3] Zuotao Zhao, Xing-Hua Gao, Yangfeng Ding, Hong Liu, Stephan Weidinger, Selena Pan, Shiqi Li, Xin Luo, Claire Feeney, Melissa Watkins, and Meena Arumugam. Efficacy of Abrocitinib and Dupilumab in Adults with Moderate-to-Severe Atopic Dermatitis Using Treat-to-Target Goals Recommended by Experts in China. *International Archives of Allergy and Immunology*, 2026. doi: 10.1159/000550965. Published online 23 February 2026.

[4] Jensen Yeung et al. Treat-to-target in the management of moderate-to-severe atopic dermatitis in adults: A canadian perspective. *Journal of the American Academy of Dermatology*, 89(2):372–375, 2023. doi: 10.1016/j.jaad.2023.01.053. URL `https://pubmed.ncbi.nlm.nih.gov/37074241/`.

[5] Jonathan I. Silverberg et al. Combining treat-to-target principles and shared decision-making: International expert consensus-based recommendations with a novel concept for minimal disease activity criteria in atopic dermatitis. *Journal of the European Academy of Dermatology and Venereology*, 38(11):2139–2148, 2024. doi: 10.1111/jdv.20229. URL `https://onlinelibrary.wiley.com/doi/10.1111/jdv.20229`.

[6] Maria Angeliki Gkini et al. Set targets, assess response, intervene early to achieve minimal disease activity in atopic dermatitis (STRIVE AD): A modified delphi consensus project to optimize management of atopic dermatitis in the united kingdom. *JAAD International*, 25:56–68, 2026. doi: 10.1016/j.jdin.2025.11.028. URL `https://pmc.ncbi.nlm.nih.gov/articles/PMC12873730/`.

[7] Joseph F. Merola et al. International Eczema Council Definitions of Low Disease Activity and Remission in Atopic Dermatitis: A Consensus Statement. *JAMA Dermatology*, 162:631–639, 2026. doi: 10.1001/jamadermatol.2026.0662.

[8] Fan Li and Fan Li. Propensity score weighting for causal inference with multiple treatments. *The Annals of Applied Statistics*, 13(4):2389–2415, 2019. doi: 10.1214/19-AOAS1282.

[9] International Council for Harmonisation. E9(r1) addendum on estimands and sensitivity analysis in clinical trials, 2019. URL `https://www.ema.europa.eu/en/documents/scientific-guideline/ich-e9-r1-addendum-estimands-and-sensitivity-analysis-clinical-trials-guideline-statistical-principles-clinical-trials-step-5_en.pdf`. Accessed 2026-10-17.

[10] J. Martin Bland and Douglas G. Altman. Agreement between methods of measurement with multiple observations per individual. *Journal of Biopharmaceutical Statistics*, 17:571–582, 2007. doi: 10.1080/10543400701329422.

[11] Judith L. Thijs et al. EASI p-EASI: Utilizing a combination of serum biomarkers offers an objective measurement tool for disease severity in atopic dermatitis patients. *Journal of Allergy and Clinical Immunology*, 140:1703–1705, 2017. doi: 10.1016/j.jaci.2017.06.046.

[12] Judith L. Thijs et al. EASI p-EASI: Predicting disease severity in atopic dermatitis patients treated with cyclosporin A. *Allergy*, 74:613–617, 2019. doi: 10.1111/all.13651.

[13] Daphne S. Bakker et al. EASI p-EASI: Predicting disease severity in atopic dermatitis patients treated with dupilumab using a combination of serum biomarkers. *Allergy*, 75:3287–3289, 2020. doi: 10.1111/all.14492.

[14] Erik von Elm et al. The Strengthening the Reporting of Observational Studies in Epidemiology (STROBE) statement: guidelines for reporting observational studies. *PLoS Medicine*, 4:e296, 2007. doi: 10.1371/journal.pmed.0040296.

[15] Gary S. Collins et al. TRIPOD+AI statement: updated guidance for reporting clinical prediction models that use regression or machine learning methods. *BMJ*, 385:e078378, 2024. doi: 10.1136/bmj-2023-078378.

[16] Karel G. M. Moons et al. PROBAST+AI: an updated quality, risk of bias, and applicability assessment tool for prediction models using regression or artificial intelligence methods. *BMJ*, 388:e082505, 2025. doi: 10.1136/bmj-2024-082505.

[17] Jonathan A. C. Sterne, Ian R. White, John B. Carlin, Michael Spratt, Patrick Royston, Michael G. Kenward, Angela M. Wood, and James R. Carpenter. Multiple imputation for missing data in epidemiological and clinical research: potential and pitfalls. *BMJ*, 338:b2393, 2009. doi: 10.1136/bmj.b2393.

[18] Rushani Wijesuriya, Margarita Moreno-Betancur, John B. Carlin, Ian R. White, Matteo Quartagno, and Katherine J. Lee. Multiple imputation for longitudinal data: A tutorial. *Statistics in Medicine*, 44(3-4):e10274, 2025. doi: 10.1002/sim.10274.

[19] Y. Wu et al. Serum biomarker-based endotypes of atopic dermatitis in China and prediction for efficacy of dupilumab. *British Journal of Dermatology*, 188:649–660, 2023. doi: 10.1093/bjd/ljad032.